

BENCHMARKING SCIENTIFIC AGENTS ACROSS DYNAMIC EARTH SYSTEMS AND NATURAL HAZARDS

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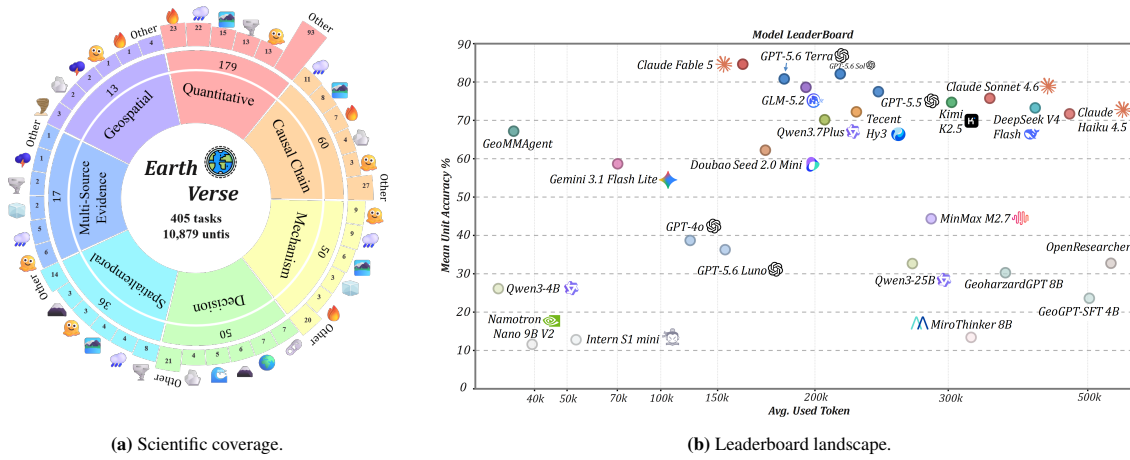


Figure 2: **Coverage and model landscape.** Task and answer-unit coverage across hazards and capabilities (left); mean answer-unit accuracy and token use across 25 systems (right).

ABSTRACT

Earth-system analysis reconstructs changing physical processes from observations that differ in source, scale, timing, and modality. Natural hazards make this work consequential because incomplete evidence can change estimates of severity, exposure, and mechanism. We introduce EarthVerse, a benchmark that evaluates scientific agents across the full investigation. Its 405 reproducible tasks are grounded in 199 documented events and 19 hazard families. Agents inspect heterogeneous event collections, choose compatible evidence, execute transparent calculations, reconcile source differences, and preserve provenance in the final account. We provide executable ground truth that decomposes each task into fine-grained answer units, together with task-specific rubrics that assess the supporting research process while allowing multiple valid paths. We evaluate 25 model and agent systems under a controlled tool-using protocol, then use controlled studies to locate failures in evidence access, tool selection, memory, reasoning, interaction, and scientific execution. Across systems, the best mean answer-unit accuracy is 84.65%, while the highest Strict@95 is only 34.81%. The gap shows that current agents often complete individual steps without maintaining a consistent chain across evidence, scales, units, calculations, and physical interpretation. EarthVerse provides a reproducible basis for measuring end-to-end scientific reliability in dynamic Earth systems.

1 INTRODUCTION

Earth-system science asks how the atmosphere, land, ocean, ecosystems, and human activity shape one another. These processes govern water, food, health, infrastructure, and climate risk. Natural hazards make the question immediate: weather, climate, and water extremes have caused extensive mortality and economic loss, while compound events can propagate across sectors and regions (World Meteorological Organization, 2023; 2025; Intergovernmental Panel on Climate Change (IPCC), 2022; Zscheischler et al., 2018). Timely analysis matters, but so does scientific traceability. A plausible explanation is not enough when it may inform disaster assessment or response.

The difficulty begins with observation. No single instrument records an evolving Earth system in full. Agency reports, stations, satellites, reanalyses, maps, and impact databases measure different variables over different footprints and time windows. Combining them requires scientific judgment before calculation: the analyst must determine which records describe the same process, which scale is appropriate, and what uncertainty remains. Work on machine learning for Earth systems and model–data integration has made these records easier to use, but has not removed the need to reconcile them (Reichstein et al., 2019; Markonis et al., 2021; Gettelman et al., 2022).

Foundation models and tool-using agents now make a broader form of analysis possible. Systems such as Google Earth AI combine imagery, population, and environmental models through a reasoning agent, while scientific agents can inspect files, execute code, and revise a result after new observations arrive (Bell et al., 2025; Yao et al., 2023; Chen et al., 2025; Hu et al., 2025; Talemi et al., 2026). This changes the evaluation target. A system has to decide what evidence to seek, bind each value to the correct source and scale, and let tool feedback alter its account when the evidence disagrees.

Most Earth-science benchmarks begin after that decision has been made. They test knowledge, figure interpretation, remote sensing, or geospatial reasoning over a supplied passage, image, or named asset (Zhang et al., 2025; Xu et al., 2026; Zhao et al., 2026). Newer multimodal and agent benchmarks broaden the interface, but generally continue to declare the observation or data layer in advance (Xiao et al., 2026; Li et al., 2025). Frontier systems already approach or exceed published expert references on some of these formats. Such results do not show whether the same system can construct a compatible evidence base for an open question, carry calculations across several sources, and preserve support for every part of its conclusion.

Position. Earth-science analysis requires systems to maintain the correspondence between heterogeneous observations and scientific claims. Satellite products, station records, reanalyses, and event reports can support a shared physical explanation only when their scales, variable meanings, time windows, units, and transformations remain aligned. EarthVerse evaluates this evolving claim–evidence state: a single broken binding can make a locally correct calculation describe the wrong Earth process (Du et al., 2025; Jiang et al., 2026).

EarthVerse contains 405 reproducible investigations grounded in 199 real disasters and extreme events. Each task opens an event collection assembled from 80 external source and access families. A typical event contains about 34 files, and solving a task requires evidence from more than seven distinct sources on average. Every file is connected to the event, so source choice remains a scientific judgment. The difficulty lies in deciding which records answer the question, whether their time windows and spatial support are compatible, and how much weight each source should carry. Working through these materials takes sustained interaction. Across the leaderboard, each task takes about 17 model rounds on average. A full evaluation of a frontier model costs more than \$2,500.

Figure 1 samples this breadth, pairing the global event collection with representative investigations that move from heterogeneous observations to an auditable conclusion.

EarthVerse places the difficulty in the research workflow. The mathematics remains transparent: aggregation, ratios, thresholds, spatial overlap, weighted indices, and bounded counterfactuals. The challenge is composition. A wrong time window changes the total; the total changes the comparison; and the comparison can change the physical explanation. Tasks therefore combine process reconstruction, quantitative analysis, competing hypotheses, rankings, and open synthesis. Fine-grained

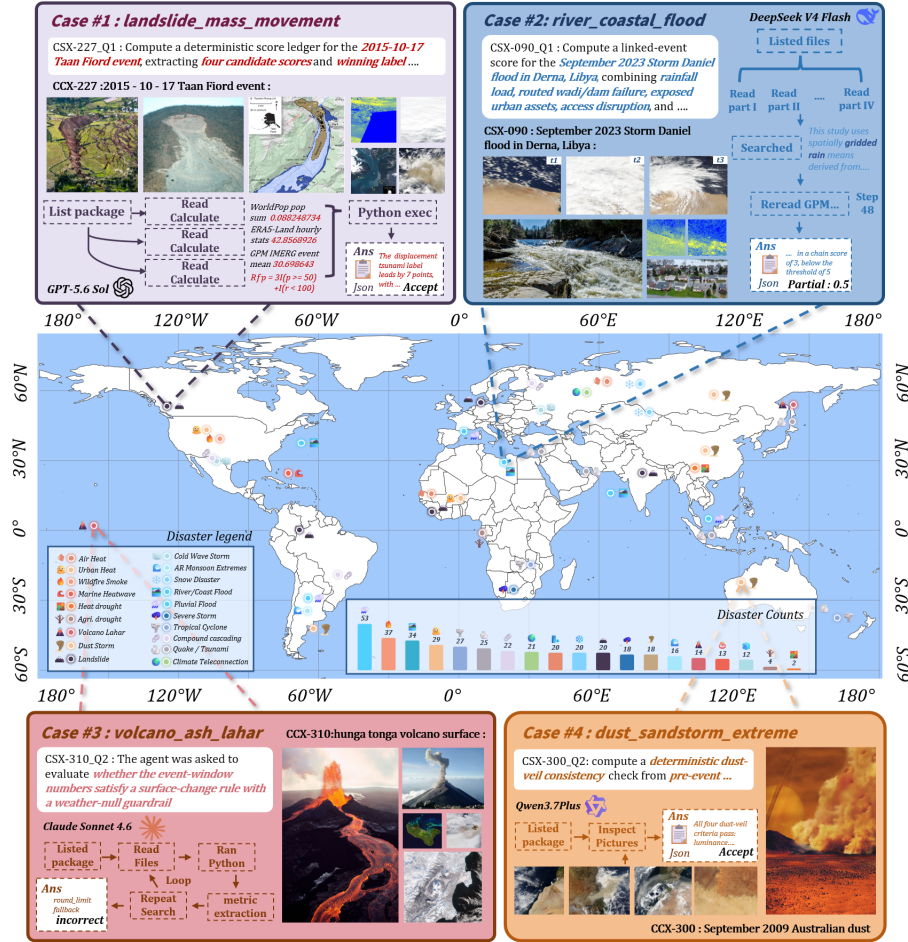


Figure 1: **Benchmark overview.** Global event coverage and representative evidence-to-answer investigations.

answer units expose each obligation, while Strict@95 asks whether the resulting account is nearly complete.

The results show how demanding this composition is. Across 25 systems, the best mean answer-unit accuracy is 84.65%, whereas the highest Strict@95 is only 34.81%. Even the strongest system leaves roughly two thirds of investigations with at least one consequential error or omission. This is the benchmark’s central reliability gap: an agent may recover most local facts and calculations while one unsupported source, mismatched scale, or missing mechanism breaks the account as a whole.

Figure 2 places the benchmark’s scientific coverage beside the resulting model landscape.

Contributions. We contribute (i) EarthVerse, a benchmark of reproducible multi-source investigations with executable answer units and task-specific process rubrics; (ii) a controlled evaluation of 25 model and agent systems that separates average scientific competence from reliable completion; and (iii) controlled experiments across evidence localization, tool selection, memory, reasoning, interaction modes, and professional scientific environments that identify the main failure points along the research chain.

Research questions. We organize the study around four questions. **RQ1:** Can current systems complete an auditable multi-source hazard investigation, and which missing links separate high average accuracy from reliable completion? **RQ2:** When does additional reasoning improve the science: over a fixed evidence snapshot, or only when the system can revise what it observes? **RQ3:**

Which controls over evidence discovery, tools, memory, stopping, and scientific execution determine whether reasoning becomes a defensible result? **RQ4:** Does performance on supplied-observation benchmarks transfer to constructing the observation set itself, and what does benchmark adaptation reveal about that gap?

2 RELATED WORK

Earth observation and scientific multimodality. EarthSE, MSEarth, GeoMMBench, AtmosSci-Bench, GeoBench-VLM, and XLRS-Bench test scientific knowledge, figure understanding, atmospheric reasoning, and remote-sensing interpretation (Xu et al., 2026; Zhao et al., 2026; Xiao et al., 2026; Li et al., 2025; Danish et al., 2025; Wang et al., 2025). Prithvi-EO-2.0 and TerraMind study representations across sensors and downstream tasks, while GEO-Bench-2 finds that no representation dominates every sensing regime (Szwarcman et al., 2026; Jakubik et al., 2025; Simumba et al., 2025). In each case, the observation or dataset is supplied.

Disaster intelligence. CrisisMMD, xBD, and FloodNet cover crisis classification, damage assessment, segmentation, and visual question answering from supplied media (Alam et al., 2018; Gupta et al., 2019; Rahnemounfar et al., 2021). DORA adds expert-authored emergency tasks, a declared multimodal manifest, typed tools, and replayable reference calls (Wang et al., 2026). It moves closer to an operational workflow, although the relevant observations and layers are still declared in advance.

Tool-using and scientific agents. ReAct interleaves reasoning and action, while Toolformer and ToolLLM study learned API use (Yao et al., 2023; Schick et al., 2023; Qin et al., 2024). AgentBench, GAIA, and OSWorld evaluate multi-step interaction in executable environments (Liu et al., 2024; Mialon et al., 2024; Xie et al., 2024). ScienceAgentBench, SciAgent, and SciAgentGym extend this setting to code, data, scientific functions, and long-horizon orchestration (Chen et al., 2025; Ma et al., 2024; Shen et al., 2026). BLADE and DiscoveryBench further expose multiple valid analysis paths and decompose data-driven discovery into evaluable facets (Gu et al., 2024; Majumder et al., 2025).

EarthVerse couples expert authoring with a scientific tool suite (Figure 3). Experts frame and verify each investigation; numerical, atmospheric, and geospatial tools support executable analysis. Evaluation then scores source choice, intermediate work, and physical interpretation separately from the final answer.

Geospatial and deep-research systems.

ThinkGeo evaluates tool execution over remote-sensing imagery; Earth-Agent, OpenEarthAgent, Spatial-Agent, and GeoMMAgent coordinate Earth-observation or geospatial operations (Shabbir et al., 2025; Feng et al., 2026; Shabbir et al., 2026; Bao et al., 2026; Xiao et al., 2026). DeepResearcher, OpenResearcher, and MiroThinker study iterative retrieval and long research trajectories (Zheng et al., 2025; Li et al., 2026; MiroMind Team et al., 2025). Intern-S1, GeoGPT, and GeohazardGPT test domain-specific pretraining (Bai et al., 2025; Zhang et al., 2024; Ge et al., 2025). EarthVerse evaluates the combined system: whether reasoning and interaction can turn retrieved evidence and tool output into a coherent account of an evolving event.

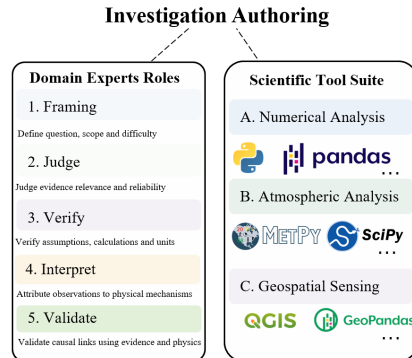


Figure 3: Expert roles and scientific tools.

3 THE EARTHVERSE BENCHMARK

EarthVerse contains 405 investigations grounded in 199 documented disasters and extreme events, covering 19 hazard families and 6,709 local files. Each task follows the rhythm of real post-event

inquiry: establish what happened, identify the observations that can resolve the question, test the relevant quantities and mechanisms, reconcile disagreement, and state what the evidence supports. Packages preserve the complexity of the event without adding large collections of unrelated files merely to obstruct retrieval.

Multi-source misalignment is part of the scientific problem. Reports, stations, satellite products, reanalyses, exposure layers, and impact records may use different windows, footprints, variables, and units. A solver must inspect these differences before choosing evidence, then manage the dependency between tools, sources, calculations, and physical reasoning. Computation is frequent but deliberately transparent: aggregation, normalization, ratios, thresholds, lags, spatial overlap, weighted indices, and bounded counterfactuals. EarthVerse does not test recall of undocumented formulas or background-heavy differential equations. It tests whether simple, checkable operations can support a complex investigation.

3.1 DESIGN PRINCIPLES

Four rules govern task construction.

Event-scale scientific iteration. The solver first identifies the event window, geographic extent, hazard evolution, evidence families, and major inconsistencies. It then alternates between targeted evidence access, calculation, and physical interpretation, revising the event picture when an observation conflicts with the current account.

Evidence discovery. The prompt names the event and scientific decision but not the relevant files. A released task requires 7.26 distinct sources on average, selected from a package with a median of 34 candidates. Across the release, evidence comes from dozens of distinct product and API families rather than one fixed archive. Tasks that reveal a path, filename, or answer-identifying quantity are rejected.

Scientific checkability. Conclusions are scored with their supporting values, units, source roles, and mechanism tests. The benchmark tests whether an operation is attached to the right evidence, not whether specialized mathematics can be reproduced from memory.

Path flexibility under package grounding. Report-first and time-series-first investigations can both receive full credit if they establish the required quantities and mechanism. Versioned packages keep the evidence and computation reproducible.

3.2 FIDELITY TO INVESTIGATIVE PRACTICE

Post-event studies usually begin with an archive rather than a curated figure. Analysts compare agency bulletins, station records at different aggregations, satellite and reanalysis products, impact databases, and incomplete metadata. One source may refine the time window; another may challenge the assumed mechanism; a third may reveal a spatial mismatch. EarthVerse reproduces this investigative sequence in a fixed, replayable setting. The scientific question defines the decision, answer units specify what must be established, and the rubric scores source choice, calculation, revision, and causal argument. The benchmark stays close to disaster practice without relying on live feeds or unstated specialist knowledge.

3.3 TASK CONSTRUCTION AND ANSWER CONTRACTS

Each event contributes two tasks, with a third for seven flagship events. The public prompt states a scientific decision but does not identify the relevant files. Every task includes an expert solution, structured ground truth, a deterministic `compute_gt.py` program, and a task-specific process rubric.

The task and evaluation flow is

$$(q_i, P_i) \xrightarrow{\pi} \tau_i = (a_1, o_1, \dots, a_T, o_T) \longrightarrow \hat{y}_i, \quad (1)$$

$$E_i = [f(\hat{y}_i, y_i^*), g(\tau_i; s_i, B_i)].$$

For task i , the solver sees question q_i and evidence collection P_i . Policy π produces trajectory τ_i and answer $\hat{y}_i$. Function f compares the answer with structured ground truth y_i^* ; g evaluates the trajectory against expert analysis s_i and rubric B_i without prescribing a tool order. We use units because a single choice cannot show whether a system recovered the correct window, ranking, quantity, mechanism, or qualification. Each unit is a nontrivial scientific obligation, and tasks can combine numerical results with orderings and open-ended conclusions. This remains discriminative when frontier models already perform extremely well on simpler multiple-choice and short-answer formats.

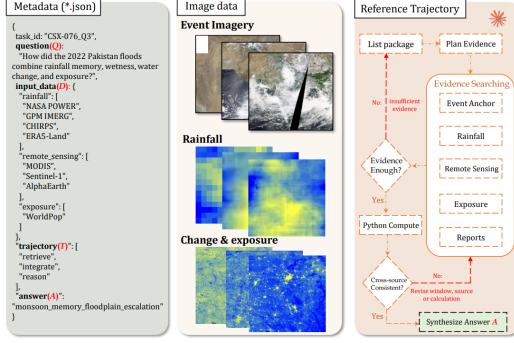


Figure 4: **Task interface.** Metadata, observations, and an auditable research trajectory.

A worked example. One task examines the 2021 Pacific Northwest heat wave. The solver must distinguish a direct hourly record from a lower-resolution aggregate, identify the matching heat window, and compute heat and humidity statistics on the same temporal support. The arithmetic is elementary; the scientific work lies in choosing compatible observations and carrying that choice into the conclusion.

3.4 CAPABILITIES AND RESEARCH ENVIRONMENT

Capability coverage. Each task receives one to three manually reviewed capability labels: physical-mechanism reasoning, spatiotemporal reconstruction, quantitative calculation, multi-source synthesis, causal-chain reasoning, ranking or decision, and remote-sensing or geospatial interpretation. Labels may overlap. Conditioned scores are diagnostic and do not change the overall metric.

Figure 2(a) shows that these capabilities recur across hazard families rather than forming separate task silos. Figure 4 illustrates how a solver links task metadata, observations, and its research trajectory.

Research tools. We selected six general operations that are sufficient to express every released workflow: discover files, read a source, search local text, inspect structured data, execute scoped Python, and finalize the answer. This common interface makes system comparisons consistent while leaving the scientific strategy open. The registry also supports custom tool registration and retains 170 reusable operations for reports, time series, vector and exposure analysis, raster and Earth observation, meteorology and climate, fire and geophysics, infrastructure, and output assembly. Structured returns separate execution failure from scientific disagreement.

3.5 QUALITY ASSURANCE AND BENCHMARK COMPARISON

Construction and review. Construction proceeds through event verification, package assembly, task authoring, executable validation, and release review (Figure 5). Automated tools support the process, but every released task is checked and corrected by people. Review covers event identity, source reliability, temporal and spatial fit, the public question, expert solution, ground truth, executable program, capability labels, and rubric. Package-local data must regenerate every deterministic value, and the visible assignment must match the hidden scoring target. Appendix C.6 describes the human and agent review process; Appendix C.7 records the seven release gates.

Position relative to prior benchmarks. Table 1 compares what the model is asked to do, what evidence it receives, how experts participate, and what is scored. EarthSE, MSEarth, and GeoMM-Bench usually provide one selected passage, figure, or image (Xu et al., 2026; Zhao et al., 2026; Xiao et al., 2026). ThinkGeo, Earth-Agent, OpenEarthAgent, and DORA add tools around declared imagery, assets, or layers (Shabbir et al., 2025; Feng et al., 2026; Shabbir et al., 2026; Wang et al.,

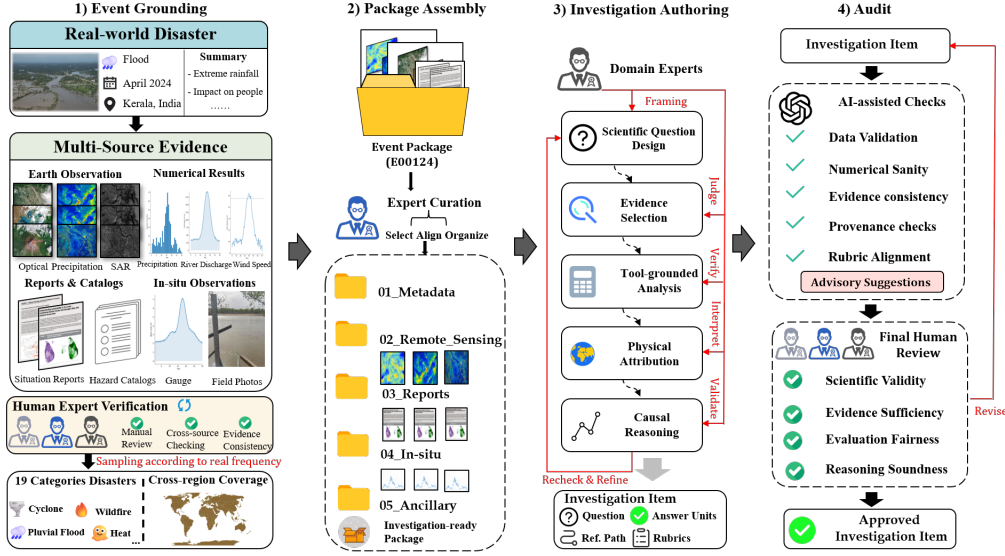


Figure 5: **Benchmark construction.** Event grounding, package assembly, investigation authoring, and expert audit.

2026). EarthVerse instead asks the system to construct and document a multi-source evidence base whose contents depend on the disaster and the scientific question.

Table 1: **Where the scientific work begins.** The two text columns identify the starting evidence and analytical work; marks show full, partial, or absent/unreported coverage.

Benchmark	Start evidence	Scientific work	Tools	Find evidence	Multi-source	Compute	Unit score	Trace score	Human review
EarthSE	Paper	Sci. QA	×	×	×	●	×	×	●
MSEarth	Figure	Figure QA	×	×	×	×	×	×	●
GeoMMBench	Image	Geo reasoning	×	×	×	×	×	×	●
ThinkGeo	RS image	Tool use	✓	●	●	✓	×	✓	●
Earth-Agent	EO assets	EO planning	✓	●	✓	✓	×	✓	●
OpenEarthAgent	Imagery	Multimodal	✓	●	●	✓	×	✓	×
DORA	Layers	Crisis analysis	✓	●	✓	✓	×	✓	●
EarthVerse	Package	Event synthesis	✓	✓	✓	✓	✓	✓	✓

✓ full ● partial × absent/unreported

EarthVerse scale: 34 files/package; 7.26 sources and 26.86 units/task.

EarthVerse places the difficulty in scientific assembly rather than specialized mathematics. Fine-grained answer units state what must be established; the process rubric tests whether the chosen path supports those claims.

4 EVALUATION PROTOCOL

EarthVerse fixes the scientific question, visible evidence boundary, execution budget, and answer contract across systems. During a run, the system selects evidence, performs calculations, records provenance, and submits a structured scientific answer through the same interface. The full inference configuration, model modalities, and controlled-study manifests are reported in Appendix C.

4.1 OUTCOME, PROCESS, AND RELIABILITY METRICS

For task i , let U_i be the percentage of required answer units satisfied and H_i the holistic answer judgment, both on a 0–100 scale. Answer correctness is $A_i = \min(H_i, U_i)$: a fluent answer cannot outrank the scientific obligations it actually completes. Let P_i be the task-specific process-rubric

score on the same scale. The primary score is

$$\text{Combined Core} = \frac{1}{2N} \sum_{i=1}^N (A_i + P_i). \quad (2)$$

It therefore gives equal weight to endpoint correctness and the documented research process.

Mean unit accuracy averages U_i across tasks and captures partial completion of required scientific claims and output fields; numerical units use task-specific tolerances. Strict@95 is the percentage of tasks with $U_i \geq 95$ and measures near-complete task reliability rather than average progress. Capability-conditioned scores apply the same Core calculation to expert-tagged task slices. Trajectory statistics report interaction, evidence access, computation, latency, and token demand, but never enter the score. Solver failures receive zero; judge-service failures are rerun until a valid judgment is obtained.

Path pluralism and diagnostic validity. EarthVerse does not prescribe a tool order. Earth-Agent and DORA instead evaluate whether an agent follows a predefined tool or operational sequence. This is useful when a workflow has a canonical procedure, but complex multi-source investigations rarely admit a single defensible trajectory ground truth: report-first and time-series-first analyses may reach the same well-supported conclusion. EarthVerse therefore scores what the system establishes and whether the observed evidence and calculations support it. Tools and traces serve reasoning, review, and diagnosis; reproducing a prescribed path earns no credit by itself.

4.2 EVALUATED SYSTEMS

Table 2 compares hosted systems, open-weight models, and research frameworks through one package interface. Some backbones are text-only, while others support native multimodal input; the shared leaderboard does not grant extra evidence to either group. All systems receive the same question, package boundary, answer contract, and execution budget. Appendix C.1 gives the complete roster, modality treatment, controller exceptions, and judge configuration.

5 MAIN EVALUATION

For RQ1, the leaderboard separates average unit competence from end-to-end completion and traces which omissions break an otherwise plausible analysis. For RQ4, cross-benchmark adaptation separates recoverable interpretation and interface errors from the harder task of constructing the evidence set. Appendix B.1 develops the resulting position.

5.1 OVERALL SYSTEM PERFORMANCE

Setting. Table 2 reports all 25 systems under the package-scoped interactive protocol. The Harness column names the controller used to run each model. Core averages Answer correctness and Process-rubric score; Strict@95 counts tasks satisfying at least 95% of answer units. Trajectory columns are per-task means and do not affect quality scores. Systems are grouped by family and ordered from lower to higher Core.

The landscape in Figure 2(b) shows that similar token budgets can produce sharply different answer-unit accuracy.

Mean competence versus conjunctive reliability. Claude Fable 5 leads on Core (84.97), while GPT-5.6 Sol leads on Strict@95 (34.81%). Fable satisfies 84.65% of answer units on average but reaches 95% completion on only 25.93% of tasks; Sol leaves fewer investigations with a consequential omission. Missing windows, provenance, alternative mechanisms, and unit conversions are coupled errors: a wrong window changes the calculation and its physical interpretation. Low Strict@95 rates show that current systems often know most of the analysis without keeping the whole analysis valid at once.

Table 2: **EarthVerse leaderboard.** Harness names the controller. Scores are percentages; failed, timed-out, or malformed runs remain zero, and batch reads may cover multiple files. Within each family, Core increases downward; **bold blue** and underlined teal mark the top two outcomes.

System configuration		Outcome					Trajectory mean per task						
System	Harness	Answer	Process	Core	Strict@95 (%)	Unit acc.	Rounds	Tool calls	Reads	Files	Python	Latency (s)	Tokens (k)
<i>Agent frameworks (own controllers over the same package interface)</i>													
OpenResearcher	OpenResearcher	30.51	31.06	30.79	4.94	32.74	39.73	33.62	15.00	17.03	5.38	521	538.7
GPT-5.5	GeoMMAgent	65.86	70.84	68.35	12.35	67.18	5.00	8.28	11.93	8.59	0.99	221	37.9
<i>Open-weight and Earth-specialized models</i>													
Intern-S1-mini	EarthVerse	10.78	12.67	11.72	0.25	11.60	4.99	3.43	1.80	2.39	0.07	57	39.4
Nemotron Nano	EarthVerse	12.30	14.01	13.15	0.25	12.80	2.59	2.07	1.26	1.77	0.15	279	53.2
MiroThinker 8B	EarthVerse	12.79	13.62	13.21	2.96	13.42	27.30	25.98	6.07	11.42	2.00	367	318.0
GeoGPT	EarthVerse	22.15	27.32	24.74	0.25	23.61	30.41	29.37	19.33	11.86	2.65	565	502.1
Qwen3-4B	EarthVerse	25.11	24.94	25.02	0.00	26.13	5.06	3.79	2.55	3.66	0.07	132	37.7
GeohazardGPT	EarthVerse	28.88	33.33	31.10	0.49	30.27	30.33	29.34	13.38	15.70	8.44	456	333.7
Qwen3-235B	EarthVerse	31.98	34.64	33.31	0.99	32.66	13.47	12.80	5.33	7.41	3.50	195	274.3
<i>Hosted general-purpose systems</i>													
GPT-5.6 Luna	Codex	35.68	35.63	35.66	11.11	36.28	6.58	5.39	2.37	4.29	1.05	189	156.9
GPT-4o	EarthVerse	37.35	41.11	39.23	0.49	38.69	18.81	8.95	4.04	6.84	0.69	55	127.4
MiniMax M2.7	EarthVerse	43.66	45.77	44.71	7.41	44.35	17.27	16.08	7.11	8.94	2.22	205	285.8
Gemini 3.1 Flash Lite	EarthVerse	57.67	61.66	59.66	2.96	58.67	7.37	6.10	3.59	5.69	1.08	24	70.1
Doubao Seed 2.0 Mini	EarthVerse	61.14	64.84	62.99	5.93	62.22	13.71	12.70	7.21	8.86	1.08	113	175.1
Qwen 3.7 Plus	EarthVerse	69.33	71.37	70.35	18.27	70.16	17.15	15.72	6.51	11.25	3.31	313	204.3
Claude Haiku 4.5	EarthVerse	70.76	73.70	72.23	14.07	71.70	21.31	19.86	10.08	11.72	3.96	324	477.0
Tencent Hy3	EarthVerse	71.27	74.26	72.77	16.30	72.22	19.47	17.77	8.32	12.70	4.41	311	208.7
DeepSeek V4 Flash	EarthVerse	72.32	74.77	73.54	20.25	73.25	27.92	26.81	11.00	16.38	4.07	333	389.7
Kimi K2.5	EarthVerse	73.89	76.76	75.32	14.57	74.70	24.25	22.67	10.01	16.00	4.79	439	302.9
Claude Sonnet 4.6	EarthVerse	74.88	77.44	76.16	20.00	75.76	17.54	16.54	7.72	11.82	4.49	343	331.0
GPT-5.5	EarthVerse	76.85	79.28	78.06	17.04	77.48	15.17	14.20	6.83	10.92	3.74	210	224.9
GLM-5.2	EarthVerse	78.14	80.19	79.17	22.96	78.62	18.32	17.27	7.52	13.36	4.31	259	196.1
GPT-5.6 Terra	Codex	80.12	80.00	80.06	<u>27.16</u>	80.81	12.18	11.13	4.42	9.46	2.71	173	182.6
GPT-5.6 Sol	Codex	<u>81.43</u>	<u>81.94</u>	<u>81.68</u>	34.81	<u>82.12</u>	13.88	12.79	4.88	9.65	2.87	205	205.9
Claude Fable 5	Claude Code	83.99	85.95	84.97	25.93	84.65	14.45	13.45	5.92	10.02	2.61	247	161.6

Finding. Reliability requires the whole evidence chain. High average accuracy can hide one missing source, window, unit, calculation, or mechanism. In disaster analysis, that single break can invalidate an otherwise plausible account.

Coupling reasoning with interaction. Most leaderboard systems use the EarthVerse harness, a common multi-round controller configured for evidence discovery, calculation, provenance, recovery, and structured finalization. Holding this scaffold fixed makes backbone comparisons fair, while framework rows retain their native controllers. Interaction volume still does not predict quality. OpenResearcher uses 39.73 rounds and 538.7k tokens per task, yet trails Claude Fable 5 by 54.18 Core points because weak sources and stale values survive into late revisions. GeoMMAgent is too brief in the opposite direction: five rounds leave evidence undiscovered and Strict@95 reaches only 12.35%. Useful interaction changes the working account and carries that change into the answer.

Finding. Useful interaction changes the scientific state. A tool call matters when its observation changes the chosen source, calculation, or physical account. Longer traces that preserve the same mistake add cost, not quality.

Domain knowledge and frontier reasoning. Earth-specialized pretraining improves terminology and process recognition (Bai et al., 2025; Zhang et al., 2024; Ge et al., 2025), but the evaluated open models remain 40–70 Core points behind frontier general systems. GeohazardGPT averages 30.33 rounds and 8.44 Python calls per task yet reaches 31.10 Core. The gap appears after recognition, in source selection, provenance, revision, and stopping. Domain training remains useful, but in these systems it has not yet produced a reliable research loop.

Capability slices in Appendix D.1 sharpen this result. Several systems rise above their own mean on calculation, spatiotemporal reconstruction, or remote sensing, yet fall on causal reasoning and ranking. Overall strength does not produce a uniform scientific profile.

Computational cost. Evaluating a frontier model costs more than \$2,500, while flash models remain below \$45. GPT-5.6 Terra retains 98% of GPT-5.6 Sol’s Core at less than half the cost. Cost depends on both model pricing and controller behavior, so neither trace length nor token count alone is an efficiency measure. Terra offers the strongest quality–cost balance among the top systems, whereas low-cost Flash models are useful for broad screening but do not match frontier reliability. Appendix D.3.3 reports the full comparison.

Where the research loop fails. Tables 3 and 4 examine the research loop from complementary directions. The first varies how a system can use evidence while it reasons; the second changes what the system can locate or act on. Read together, they separate longer deliberation from revisions that materially improve the scientific account.

Table 3: Reasoning and evidence access. Complete GPT-5.5 sweep across four access modes and five effort settings. Interaction and evidence diagnostics apply to interactive runs; total tokens and latency are per-task means.

Effort	Outcome				Interaction		Evidence		Resources	
	Answer	Process	Core	Unit acc.	Rounds	Calls	Files	Refs	Tokens (k)	Latency (s)
<i>Compressed direct</i>										
none	41.37	45.10	43.24	42.21	–	–	–	–	13.9	55.6
low	24.99	29.32	27.16	25.43	–	–	–	–	14.9	120.5
medium	22.88	25.69	24.29	23.26	–	–	–	–	15.8	152.4
high	24.17	28.17	26.17	24.28	–	–	–	–	16.6	71.5
xhigh	16.67	16.67	16.67	16.67	–	–	–	–	17.6	88.1
<i>All-text direct</i>										
none	27.66	31.22	29.44	28.27	–	–	–	–	81.6	32.1
low	13.26	16.20	14.73	13.36	–	–	–	–	82.7	44.1
medium	9.79	12.18	10.99	10.05	–	–	–	–	83.5	61.0
high	8.80	9.00	8.90	8.82	–	–	–	–	85.4	78.5
xhigh	12.00	15.00	13.50	12.00	–	–	–	–	87.7	94.4
<i>EarthVerse interactive</i>										
none	48.89	53.70	51.30	49.46	9.68	8.68	7.00	9.94	96.9	140
low	56.97	60.05	58.51	57.53	14.70	13.70	9.44	14.70	162.4	259
medium	61.45	64.80	63.13	62.35	17.20	16.20	10.71	18.12	206.4	424
high	72.41	75.19	73.80	72.59	14.96	13.96	9.92	17.19	174.3	571
xhigh	69.66	67.50	68.58	69.83	11.25	10.25	10.25	16.50	100.0	674
<i>Interactive with final review</i>										
none	53.70	56.48	55.09	54.06	11.62	9.62	7.42	10.46	130.7	108
low	61.93	65.50	63.72	62.78	18.18	16.18	10.60	17.02	227.0	253
medium	67.16	69.22	68.19	67.91	20.81	18.81	11.88	21.02	266.3	552
high	69.82	74.71	72.27	70.73	20.63	18.63	12.13	21.10	255.1	1,105
xhigh	84.36	88.21	86.29	84.40	17.57	15.57	10.14	18.29	217.7	1,493

Table 3 shows a sharp split between fixed and editable evidence. With compressed or all-text direct input, the best result comes from no additional reasoning; high effort lowers Core from 43.24 to 26.17 and from 29.44 to 8.90, respectively. Once the system can retrieve evidence and act on what it finds, Core rises from 51.30 with no extra effort to 73.80 at high effort. Final review changes the best operating point again: xhigh reaches 86.29, compared with 68.58 without review. Reasoning helps when it can trigger retrieval, recomputation, or correction. Deliberating longer over the same snapshot often deepens the initial mistake.

Finding. Reasoning needs an editable evidence state. More reasoning is useful only when it can change the investigation. Interactive retrieval and final review let the system replace a weak source, recompute a quantity, and carry the correction into the final account. On a fixed snapshot, the same extra effort can reinforce an early error.

Table 4 identifies what changes that investigation. Relevant-file localization and the evidence map raise Core by 14.72 and 11.34 points, and they are the only interventions whose confidence intervals exclude zero. A curated tool subset adds 2.19 points, while the planning oracle slightly lowers performance. Renaming tools, normalizing units, and exposing optional meteorological tools also fail to reproduce the localization gain. More tools or more procedural guidance are therefore not enough. The agent must find the observations that determine the scientific object before it can reason reliably about that object.

The capability columns make the mechanism more precise. Relevant-file localization lifts spatiotemporal Core from 64.40 to 91.47, physical-mechanism Core from 65.15 to 80.61, and quantitative-calculation Core from 73.43 to 87.57. Multi-source performance changes little because the baseline already covers many sources; the gain comes from selecting and aligning the right ones. Missing,

Table 4: **Evidence-path interventions.** Complete paired outcome, effect, and capability grid relative to the Top-20 router. Intervals are paired 95% bootstrap CIs.

Intervention	Outcome			Paired effect		Core by capability						
	Answer	Process	Core	Δ Core	95% CI	Physical mech.	Spatio-temporal	Quant. calc.	Multi-source	Causal chain	Ranking	RS / geo.
Top-20 semantic-router baseline	66.90	65.88	66.39	0.00	–	65.15	64.40	73.43	90.00	65.89	50.84	53.94
<i>Answer-free information oracles</i>												
Relevant-file oracle	79.46	82.75	81.11	+14.72	[+5.48, +24.69]	80.61	91.47	87.57	90.45	76.53	60.70	72.09
Evidence-map oracle	76.96	78.50	77.73	+11.34	[+2.34, +21.25]	76.69	90.50	83.83	97.50	73.35	58.16	61.25
Tool-subset oracle	68.53	68.62	68.58	+2.19	[−3.55, +7.28]	67.32	65.67	72.60	92.50	69.54	59.14	59.75
Planning oracle	65.19	66.50	65.84	−0.54	[−7.07, +4.35]	64.31	57.10	71.24	95.00	67.93	54.76	60.25
<i>Tool-interface presentation</i>												
Hierarchical catalog	66.76	67.62	67.19	+0.80	[−9.18, +11.86]	65.81	66.72	67.83	93.50	70.78	56.60	61.67
Descriptive tool names	66.88	66.75	66.82	+0.43	[−3.55, +4.05]	65.07	61.21	72.92	100.00	66.98	57.59	55.23
Dynamic descriptions	61.39	61.75	61.57	−4.82	[−15.02, +4.33]	59.81	61.56	63.81	95.00	61.27	54.58	60.75
Opaque tool names	56.72	59.50	58.11	−8.28	[−23.57, +6.33]	56.16	48.88	58.54	95.22	59.76	54.01	80.00
<i>Evidence perturbations</i>												
Equivalent unit change	67.61	71.12	69.37	+2.98	[−9.75, +15.53]	68.02	66.93	71.44	95.00	71.29	55.47	76.38
Added distractor files	65.91	67.75	66.83	+0.44	[−6.80, +7.27]	65.46	60.47	72.15	92.95	65.16	59.98	70.62
Missing evidence	63.97	64.75	64.36	−2.03	[−12.05, +7.08]	62.49	60.31	67.82	100.00	66.63	54.74	55.12
Truncated evidence	63.17	63.38	63.27	−3.12	[−14.13, +7.47]	61.78	55.67	70.53	91.70	64.24	41.52	68.92
Conflicting evidence	63.03	63.12	63.08	−3.31	[−12.05, +2.97]	61.50	67.97	66.63	92.95	60.46	55.19	54.50
<i>Optional meteorological access</i>												
Compact meteorology suite	62.69	65.38	64.03	−2.36	[−11.92, +5.89]	62.14	61.97	66.67	100.00	65.22	56.15	58.47
Cue-routed meteorology	57.22	59.88	58.55	−7.84	[−19.40, +3.93]	56.74	58.87	59.25	92.95	60.28	50.45	57.25

truncated, or conflicting evidence lowers overall quality, although the paired intervals remain wide. In this study, evidence localization is the clearest reproducible bottleneck, and downstream reasoning quality depends on whether that step succeeds.

5.2 CROSS-BENCHMARK TRANSFER AND BENCHMARK ADAPTATION

We evaluate GPT-5.5 under the public protocols of Earth-Bench, EarthSE, MSEarth, and GeoMM-Bench. Each comparison begins with a direct response, applies a corrected stage that repairs clearly erroneous benchmark items or judgments, and then applies the structured EarthVerse procedure derived from our task design. Table 5 retains each benchmark’s native metric, the strongest relevant published line, and human references for MSEarth and GeoMMBench (Zhao et al., 2026; Xiao et al., 2026). Full source-paper rows are reported with the release artifacts.

Table 5: **Protocol sensitivity across Earth-science benchmarks.** Each panel retains its native metric and the rows needed to compare the published reference, direct GPT-5.5, the corrected stage, and the EarthVerse procedure. Published values are marked *; **bold blue** marks panel bests.

Earth-Bench			EarthSE Iron				
Model / stage	Agent	Instr.	Model / stage	MCQ	T/F	Fill	Free
Published best*	61.00	64.00	Published best*	94.01	88.64	21.85	98.70
GPT-5.5 / direct	73.49	78.31	GPT-5.5 / direct	95.48	79.85	20.29	98.00
GPT-5.5 / corrected	83.73	81.93	GPT-5.5 / corrected	95.60	80.22	58.13	99.35
GPT-5.5 / EarthVerse	87.35	84.94	GPT-5.5 / EarthVerse	97.56	86.26	61.90	98.35
MSEarth			GeoMMBench				
Model / stage	Full set	Mini recon.	Model / stage	Accuracy			
Published best*	68.82	–	Human experts*	86.50			
Human experts*	–	87.00	GeoMMAgent*	88.40			
GPT-5.5 / direct	67.60	63.00	GPT-5.5 / direct	84.52			
GPT-5.5 / corrected	68.28	86.00	GPT-5.5 / corrected	93.73			
GPT-5.5 / EarthVerse	79.96	89.67	GPT-5.5 / EarthVerse	98.01			

Structured adaptation resolves benchmark interfaces. Direct GPT-5.5 trails the strongest published result on MSEarth and GeoMMBench. The EarthVerse stage closes both gaps and raises EarthSE fill-in accuracy from 20.29 to 61.90. Most of the gain comes from resolving aliases, units, task taxonomy, and answer granularity rather than learning new Earth science during evaluation.

Fixed-observation tasks no longer separate frontier systems. The strongest closed model exceeds the displayed expert reference on MSEarth and GeoMMBench after structured adaptation. On scientific QA and figure interpretation that do not require broad evidence construction, frontier systems can already surpass the published human baselines. These benchmarks reveal useful knowledge and perception skills, but they leave much of the scientific process outside the task. EarthVerse shifts the test to constructing a compatible observation set, calculating intermediate results, and maintaining provenance across the analysis.

Finding. Evidence construction remains the hard step. The EarthVerse procedure exceeds published human reference lines on supplied-observation tasks, yet EarthVerse Strict@95 remains below 35%. Interpreting selected evidence is easier than building and validating the evidence base.

6 CONCLUSION

RQ1: scientific reliability is conjunctive. The best mean unit accuracy is 84.65%, while the highest Strict@95 is only 34.81%. Current systems usually recover much of an analysis, but a single unsupported source, mismatched scale, or missing mechanism can invalidate the result. Progress therefore requires preserving the whole claim–evidence chain, not only raising average factual accuracy.

RQ2: reasoning is useful when it changes the investigation. More effort over a fixed serialization does not recover interactive performance. Productive deliberation identifies what remains uncertain, selects the next observation or calculation accordingly, and updates the working account. Without that coupling, additional thought becomes repetition or overcommitment.

RQ3: control determines whether tools improve science. Evidence localization and selective routing help, whereas unrestricted catalogs and unstructured memory introduce competing assumptions and stale values. A reliable scientific controller must maintain claim–source bindings, check method applicability, record replaced quantities, and stop when the required evidence is complete.

RQ4: supplied-observation success does not imply scientific autonomy. Structured adaptation crosses published human reference lines when the observation is fixed, yet EarthVerse Strict@95 stays below 35%. Frontier models can interpret selected evidence extremely well; the unresolved problem is constructing a compatible evidence base and keeping it valid through calculation and synthesis.

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Reading routes

Experiments **A + D**

Reproduction **C + E**

Interpretation **B + F + G**

A CONTROLLED STUDIES

RQ2 asks whether reasoning helps without the ability to revise the observed evidence. RQ3 asks which parts of the research loop preserve or break scientific continuity. We therefore vary answer mode, reasoning effort, evidence localization, tool routing, context, stopping, and professional scientific execution. Unless noted, every condition uses GPT-5.5 under the protocol in Section 4.2, with identical task manifests within each experiment.

A.1 REASONING EFFORT AND ANSWER MODE

We cross four answer modes with five reasoning settings on a fixed, difficulty-stratified manifest. Compressed direct uses a bounded evidence summary, while all-text direct serializes every readable file. Both interactive conditions use the same EarthVerse harness; the review variant adds one evidence-and-calculation audit before finalization. The compact view below retains the outcome columns so that each reasoning curve can be compared without the denser interaction and resource diagnostics in Table 3.

Table 6: **Outcome view of the reasoning sweep.** GPT-5.5; **bold blue** and shading mark the best setting in each access mode.

Compressed direct (selected evidence)					All-text direct (budgeted evidence)				
Effort	Answer	Process	Core	Unit acc.	Effort	Answer	Process	Core	Unit acc.
none	41.37	45.10	43.24	42.21	none	27.66	31.22	29.44	28.27
low	24.99	29.32	27.16	25.43	low	13.26	16.20	14.73	13.36
medium	22.88	25.69	24.29	23.26	medium	9.79	12.18	10.99	10.05
high	24.17	28.17	26.17	24.28	high	8.80	9.00	8.90	8.82
xhigh	16.67	16.67	16.67	16.67	xhigh	12.00	15.00	13.50	12.00
EarthVerse interactive					EarthVerse interactive + review				
Effort	Answer	Process	Core	Unit acc.	Effort	Answer	Process	Core	Unit acc.
none	48.89	53.70	51.30	49.46	none	53.70	56.48	55.09	54.06
low	56.97	60.05	58.51	57.53	low	61.93	65.50	63.72	62.78
medium	61.45	64.80	63.13	62.35	medium	67.16	69.22	68.19	67.91
high	72.41	75.19	73.80	72.59	high	69.82	74.71	72.27	70.73
xhigh	69.66	67.50	68.58	69.83	xhigh	84.36	88.21	86.29	84.40

A.2 TOOLS, CONTEXT, AND INTERACTION BUDGET

On one fixed paired manifest, we vary catalog size, semantic routing, tool composition, context retention, and interaction budget. Every condition uses the default medium reasoning setting and the same answer and process scores. Appendix D.4 reports trajectory diagnostics.

A.2.1 TOOL-CATALOG SCALE

Nested fixed catalogs expose increasingly broad subsets of the same 170-tool registry (panel (a) of Table 7). The six-tool condition contains package discovery, reading, search, tabular inspection, Python execution, and finalization; each larger catalog adds domain tools without changing the task prompt.

The full catalog loses 9.80 Core points relative to the six-tool foundation while using 19% more tokens. The decline is not monotonic, which rules out a simple capacity penalty. Extra tools add nearby analytical paths with different assumptions about variables, units, aggregation, and applicability. When the controller cannot retain why it chose one path, later outputs from another tool can enter the same ledger as if they were comparable. Catalog breadth then creates scientific ambiguity. Reporting a tool count without the navigation policy misses this failure mode.

A.2.2 SEMANTIC ROUTING AND SCIENTIFIC COMPOSITION

The routing study selects tools from task text and package metadata before interaction (panel (b) of Table 7). Top- k routing exposes the highest-scoring individual tools, whereas workflow routing first selects a domain workflow. A separate composition study compares an expert-only suite, a mixed general/expert suite, and atmospheric Python and Fortran suites.

Top-20 routing recovers 11.21 Core points over the full catalog with 34% fewer tokens. Its advantage comes from narrowing the next choice without deciding the investigation in advance. Workflow routing is more brittle because it can commit to a domain before the evidence warrants that commitment; the expert-only suite has the opposite problem, lacking ordinary file and calculation operations that connect specialist output to a scored claim. Routing should act as a revisable prior: preserve a small general core, introduce specialist methods when their required variables appear, and allow the evidence to overturn the initial route.

A.2.3 CONTEXT RETENTION

We vary the amount of tool history preserved verbatim before older content is summarized (panel (c) of Table 7). Aggressive compression keeps a narrow recent window; short and long policies progressively retain more observations; full history disables summarization within the tested budget.

Long bounded retention gains 13.66 Core points over aggressive compression and 5.27 over full replay. Too little history forces the agent to rediscover files and often severs intermediate values from their provenance. Full replay fails differently: rejected hypotheses and superseded numbers remain salient, so they can reappear in later calculations. A transcript preserves everything but distinguishes nothing. Scientific memory needs an active ledger that binds claims to sources, units, temporal and spatial support, and unresolved conflicts, while archiving values that have been replaced.

Table 7: **System ablations.** GPT-5.5 at medium reasoning; **bold blue** marks panel bests.

Condition	Outcome			Coverage		Core by capability						Resource	
	Answer	Process	Core	Unit acc.	Physical mech.	Spatio-temporal	Quant. calc.	Multi-source	Causal chain	Ranking	RS / geo.	Tokens	
<i>(a) Tool catalog: nested subsets of 170 tools</i>													
Fixed tools (6)	64.34	65.62	64.98	64.69	63.27	66.22	70.34	97.50	63.35	51.78	58.12	229,557	
Fixed tools (10)	60.22	57.10	58.66	60.35	56.74	58.36	61.04	95.22	55.77	58.32	61.25	194,047	
Fixed tools (20)	60.58	59.38	59.98	60.93	58.13	70.50	61.67	95.00	54.90	56.50	55.62	232,912	
Fixed tools (40)	58.55	61.00	59.77	59.43	58.16	47.81	62.03	90.45	63.63	59.57	54.12	232,993	
Fixed tools (70)	58.76	61.25	60.01	59.19	58.54	55.68	63.89	87.95	60.14	54.21	54.62	233,759	
Fixed tools (100)	63.96	63.12	63.54	64.11	61.81	58.93	65.84	96.37	65.79	60.82	51.00	278,936	
Full catalog (170)	54.48	55.88	55.18	55.46	55.58	55.72	54.96	47.50	52.75	63.15	55.00	273,216	
<i>(b) Routing and tool composition</i>													
Top-10 semantic router	64.12	66.25	65.19	64.55	64.37	60.64	67.93	80.74	67.50	57.20	59.17	216,227	
Top-20 semantic router	66.90	65.88	66.39	67.14	65.15	64.40	73.43	90.00	65.89	50.84	53.94	180,010	
Workflow-profile router	56.05	56.75	56.40	56.47	54.74	52.63	58.60	87.95	55.39	55.00	60.62	220,232	
Expert-only suite	59.28	60.88	60.08	59.50	58.48	61.08	61.71	90.45	57.79	61.08	57.62	295,490	
General + expert suite	59.07	63.00	61.04	59.95	59.25	66.01	63.21	95.00	61.34	49.27	51.38	195,092	
Atmospheric Python suite	59.48	60.12	59.80	59.67	58.26	59.64	60.69	89.20	60.94	58.31	49.25	198,802	
Atmospheric Fortran suite	57.46	57.88	57.67	57.51	55.81	51.05	61.19	92.95	57.35	53.88	60.75	205,262	
<i>(c) Context retained before summarization</i>													
Aggressive compression	48.75	51.62	50.19	49.00	48.07	57.55	53.73	90.45	44.65	45.44	47.00	197,038	
Short retained context	56.31	56.38	56.34	56.52	54.16	57.30	55.54	97.72	57.85	60.08	42.25	209,827	
Long retained context	64.32	63.38	63.85	64.75	62.21	59.43	68.29	95.00	64.72	58.79	48.25	227,536	
Full retained history	58.90	58.25	58.58	59.42	56.98	55.39	62.15	89.00	57.25	52.59	61.97	187,700	
<i>(d) EarthVerse-harness turn budget under one wall clock</i>													
12-turn budget	44.84	49.50	47.17	44.95	44.65	45.36	46.27	95.00	47.73	57.50	35.13	106,263	
24-turn budget	54.10	52.88	53.49	54.32	51.31	57.88	55.46	95.00	50.93	49.13	50.92	161,404	
96-turn budget	59.72	60.12	59.92	60.16	57.94	62.76	60.53	97.50	62.10	54.08	44.38	182,798	
Wall-clock limited	62.65	64.50	63.57	62.91	62.16	62.72	66.66	90.45	67.11	56.91	33.38	221,180	

Finding. Scientific memory is selective. Long bounded retention beats both aggressive compression and full replay. The useful state is a compact ledger of active claims, sources, units, and unresolved conflicts, not a verbatim transcript.

A.2.4 INTERACTION BUDGET

The final study changes only the EarthVerse-harness turn limit. The unbounded condition still uses the common wall-clock budget, so it tests productive stopping rather than unlimited execution.

Twelve turns usually end before discovery, calculation, and verification can all occur. Additional turns improve mean quality, but the gain depends on what those turns accomplish. Extra interaction helps when it closes a named evidence gap and hurts when the agent reopens a settled calculation or follows an unproductive branch. Stopping is therefore a coverage decision: the system should stop when required claims are supported, units and provenance are checked, and material contradictions have been resolved or disclosed.

A.3 ORACLE ACCESS, DISCOVERABILITY, AND EVIDENCE ROBUSTNESS

These conditions reuse one paired task set and the Top-20 router baseline. Table 4 reports the outcomes; Appendix D.5 retains trajectory and resource diagnostics.

A.3.1 ORACLE BOTTLENECK LOCALIZATION

The relevant-file oracle reveals only the package-relative files needed by the reference solution (panel (a) of Table 4). The tool-subset oracle reveals useful tool classes, the planning oracle supplies an answer-free high-level plan, and the evidence-map oracle supplies evidence roles and intermediate values without revealing the final conclusion.

A.3.2 TOOL NAMING AND DISCOVERABILITY

We hold tool implementations fixed and alter only how the catalog is presented (panel (b) of Table 4). Descriptive names add intended-use and exclusion cues; hierarchical discovery chooses a domain before exposing tools; dynamic descriptions retrieve candidate tools at each step; opaque names remove semantic hints.

A.3.3 EVIDENCE PERTURBATION AND ROBUSTNESS

Perturbations preserve the question and answer schema while modifying the package evidence (panel (c) of Table 4): irrelevant files are added, a secondary source is made inconsistent, one supporting file is removed, long text is truncated, or units are changed while preserving physical equivalence.

A.3.4 COMPACT METEOROLOGICAL ACCESS

The final probe compares a fixed compact meteorological suite with a cue-based router that exposes at most three meteorological operations (panel (d) of Table 4). Unlike the professional-environment study below, these conditions do not require use of the specialist tools.

A.4 PROFESSIONAL METEOROLOGICAL ENVIRONMENTS

We require repeated use of one of 13 professional libraries or runtimes under the same protocol. Each adapter exposes only implemented operations and records execution success, so the comparison reflects both scientific fit and interface reliability.

Julia, R climate, MetPy, xclim, and CDO improve Core because their implemented operations match common package structures such as time series, gridded fields, and atmospheric profiles. Julia gives the largest gain (+4.62), but the ranking is less informative than the conditions under which the tools help. They compress many low-level numerical steps into an inspectable transformation without hiding the variables or aggregation. That advantage disappears when the package lacks the temporal support or paired variables assumed by the method.

Table 8: **Required scientific environments.** GPT-5.5; Δ Core is relative to the Top-20 router.

Environment	Outcome			Change	Coverage
	Answer	Process	Core	Δ Core	Unit acc.
Top-20 router (ref.)	66.90	65.88	66.39	0.00	67.14
<i>(a) Python scientific libraries</i>					
MetPy	68.69	70.38	69.53	+3.14	68.80
xclim	67.77	69.62	68.70	+2.31	67.95
thermofeel	66.78	68.75	67.76	+1.37	67.47
PyET	65.34	67.38	66.36	−0.03	65.89
climate-indices	64.50	66.75	65.62	−0.77	65.58
xskillscore	63.37	64.12	63.75	−2.64	63.90
pyextremes	59.02	60.00	59.51	−6.88	59.27
<i>(b) Compiled and domain-language environments</i>					
Julia	71.77	70.25	71.01	+4.62	71.99
R climate	68.55	71.38	69.96	+3.57	69.44
CDO	66.91	69.50	68.20	+1.81	67.35
NCO	67.39	65.00	66.19	−0.20	67.49
NCL	65.03	65.00	65.01	−1.38	65.27
Fortran	64.46	64.25	64.35	−2.04	64.52

Required use exposes a failure that optional-tool studies can miss. Pyextremes loses 6.88 Core points because many event packages do not contain a long, homogeneous series suitable for return-level estimation; forecast-verification methods face the same mismatch when paired forecasts and observations are absent. Under a hard requirement, the agent sometimes bends the available data toward the method instead of rejecting the method. A scientific interface needs an explicit refusal path, backed by minimum record length, sampling assumptions, missing-data rules, and a statement of what the output can support.

Finding. Execution success is not scientific validity. A specialist method helps only when its required variables and sampling support exist. Otherwise, a successful call can produce a precise result that the package cannot justify.

B DISCUSSION, BROADER IMPACT, AND LIMITATIONS

B.1 RESEARCH POSITION

EarthVerse evaluates a complete research loop over a bounded event archive. The agent must decide what counts as evidence, transform it, and revise an event-scale explanation while retaining enough provenance for another analyst to replay the work. This is more constrained than open-ended research, but it begins earlier than supplied-observation QA and asks more of the controller than a tool-execution benchmark.

Strong systems often interpret a selected observation correctly. Reliability falls when they must choose the record, its scale, the compatible variables, and the point at which the analysis is complete. Many failures originate in *evidential binding*: a plausible value is attached to the wrong aggregation, time window, unit, or source role, and later reasoning treats the mismatch as established fact. This explains why stronger reasoning, more tools, and longer traces can all fail to help. They operate on the state the controller has preserved. If that state no longer records what a value means, additional reasoning only makes the wrong account more elaborate.

Complex Earth-system investigations rarely have a unique valid research path. EarthVerse therefore prescribes neither a tool order nor a reference trajectory. Different paths are valid when they recover the required windows, quantities, source roles, and mechanisms, and preserve support for the resulting claims (Gu et al., 2024; Majumder et al., 2025).

B.2 BROADER IMPACT

Scientific practice. Assistants can already reduce the mechanical burden of first-pass hazard analysis: inventorying an archive, locating candidate records, harmonizing units, repeating calculations, and surfacing source disagreement. The experiments also show where this assistance becomes risky. A fluent system may complete each local operation while carrying a scale mismatch through the full analysis. Experts still need to judge comparability, uncertainty, and consequences; package-local provenance gives them something concrete to inspect.

Reliability in high-stakes settings. High average unit accuracy can hide one missing source, time window, calculation, or mechanism. In disaster analysis, that single omission may change estimated severity, exposed population, or the causal account used to justify action. The benchmark does not certify operational readiness. It makes these omissions visible before a result is treated as operational evidence.

Access and evaluation practice. A local package reduces dependence on changing APIs and lets researchers inspect the same evidence. Evaluating frontier systems across hundreds of long trajectories is still expensive and may exclude smaller groups. Cost should be reported alongside quality, and releases should include compact diagnostic subsets, task-level scores, and traces. Subset results should not be presented as full-benchmark estimates.

B.3 SYSTEM DESIGN IMPLICATIONS

For Earth-system analysis, a trajectory is useful only when it preserves the support for each claim. A record of calls without source bindings cannot show whether the final explanation follows from the evidence.

Evidence state. A transcript preserves chronology, not scientific status. The controller needs a claim–source graph in which every derived value retains its source, unit, time window, transformation, and relation to competing explanations. Corrections should replace active values without erasing the history that explains the change.

Scientific operations. Scientific tools need to expose applicability as part of their output. Record length, sampling support, missing-data rules, and estimator assumptions determine whether a computed value can support the claim. Training that rewards successful calls alone encourages the model to force data into an available method; justified refusal should receive credit when the inputs are inadequate.

Verification and oversight. Final review should operate on claims rather than prose. For each required conclusion, the system should expose coverage, dimensional checks, source conflicts, and the mechanism tests it rejected. Experts can then inspect the points where judgment entered the analysis and decide whether the result is fit for use. A polished report without this record is not auditable.

B.4 LIMITATIONS

Coverage and ecological validity. EarthVerse is a controlled approximation of research. Its 199 packages cover many natural hazards, but not the full range of Earth-system science, long-term attribution, or live forecasting. Package isolation improves reproducibility and blocks web leakage, at the cost of open-ended discovery and institutional data-quality judgments. Masked products also prevent tests of absolute geolocation.

Ground truth and judge validity. Structured ground truth necessarily chooses one decomposition of a scientific answer. Process scores also rely on an LLM judge that compares each trace with a reference solution. Separate answer and process blocks reduce score leakage but cannot remove judge error. Future releases should measure agreement with domain experts on a stratified subset and retain criterion-level decisions.

Data provenance and maintenance. Package-local provenance makes an analysis replayable, but not every source is complete or redistributable. Releases must retain manifests, access dates, transformations, permissions, checksums, and change logs. Masked products support derived statistics but not absolute-location claims, while narrative reports inherit uneven observation and reporting practices. Documented events may also appear in model pretraining. Package isolation blocks unrecorded web evidence during a run but cannot prove contamination-free recall.

Statistical scope of controlled studies. The controlled studies use paired subsets, and several intervals in Appendix D.5 include zero. Large contrasts, such as the relevant-file oracle, support architectural hypotheses. Smaller changes need replication across more packages and backbones. Cross-benchmark revision measures recoverability under feedback, not zero-shot superiority.

C EXPERIMENTAL SETUP AND REPRODUCIBILITY

This section collects implementation details that are necessary for reproduction but not for interpreting the main findings. It records the evaluated systems, the common evidence interface, the leaderboard configuration, controlled-study manifests, scoring isolation, and retained audit artifacts.

C.1 SYSTEMS, CONTROLLERS, AND MODEL MODALITY

The leaderboard contains 25 complete systems. Hosted endpoints include the OpenAI GPT-4o, GPT-5.5, and GPT-5.6 tiers (OpenAI, 2024; 2026a;b); Anthropic Claude tiers (Anthropic, 2026); Gemini (Google, 2026); MiniMax M2.7 (MiniMax, 2026); Seed 2.0 (ByteDance Seed Team, 2026); Qwen3.7 (Qwen Team, 2026); Tencent Hy3 (Tencent Hunyuan, 2026); DeepSeek-V4 (DeepSeek-AI, 2026); Kimi K2.5 (Kimi Team, 2026); and GLM-5.2 (Z.ai, 2026). The open-weight group includes Qwen3 at 4B and 235B scales (Yang et al., 2025), Nemotron Nano (NVIDIA, 2025), MiroThinker 8B (MiroMind Team et al., 2025), GeohazardGPT, GeoGPT, and Intern-S1-mini (Ge et al., 2025; Zhang et al., 2024; Bai et al., 2025).

Most systems run in the EarthVerse harness, which manages package discovery, tool use, context, recovery, and finalization. GeoMMAgent uses its adapted five-stage coordinator with GPT-5.5 as the worker model in every stage; OpenResearcher retains its research-loop controller (Xiao et al., 2026; Li et al., 2026). These rows therefore describe a model–controller system, not a bare backbone.

The roster mixes text-only models with natively multimodal endpoints. Direct image interpretation is required by only a small fraction of EarthVerse tasks; most remote-sensing evidence is represented by product metadata, raster statistics, change indices, or other auditable tool outputs. Raw pixels are not selectively sent to models with vision support. This keeps the evidence interface comparable, although the resulting scores should not be read as a standalone test of visual perception.

C.2 SHARED EXPERIMENTAL CONFIGURATION

Every run exposes a scientific question and one evidence root while keeping the solution, rubric, ground truth, scoring code, prior outputs, and web evidence hidden. Reads, writes, and Python execution stay inside the active evidence collection. Reports and tables are returned as local text or structured summaries; image and raster operations return recorded metadata and measurements.

The EarthVerse harness is the common controller for the standard model rows. To study the attainable system-level ceiling, we also evaluate the GPT-5.6 family in Codex and Claude Fable 5 in Claude Code, the strongest model-native harnesses available for those systems. GeoMMAgent pairs GPT-5.5 with its five-stage coordinator, while OpenResearcher keeps its own controller. The Harness column in Table 2 makes these differences explicit.

The common execution limit is 48 turns, 1,680 seconds for research, and 1,800 seconds overall. The controller retains 16 recent messages and a 24,000-character research state; individual tool observations are capped at 12,000 characters. Package-scoped Python has a 60-second call limit. Up to two schema repairs are allowed, with 120 seconds reserved for finalization and a separate 120-second final-call timeout. We retain provider-native reasoning and generation controls when available, use endpoint defaults otherwise, and do not impose one temperature across providers. The

judge is a fixed gpt-4.1 endpoint, and randomized procedures use seed 2026. Endpoint access dates and local model revisions are retained in the release registry.

C.3 PACKAGE EVIDENCE SOURCES AND ACCESS INTERFACES

The 199 package manifests contain 6,397 provenance records, 835 raw `source_name` labels, and 6,709 local files from 107 external host domains. Many raw labels are event-specific filenames, response formats, endpoint variants, or repeated queries to the same service rather than distinct sources. We therefore group aliases at the product/API-family level while keeping distinct providers, sensors, product versions, and access semantics separate. This yields 80 external source and access families. We also list nine package-local derivative families used directly in computation; they do not count as additional upstream sources.

PACKAGE SOURCES Distinct source and access families across 199 event packages

Event discovery, reports, and authoritative anchors

29 families

Provider or product	Access form	Role in a package
GDACS	event-list REST/GeoJSON API; RSS archive	Cross-hazard event discovery, dates, alert levels, and identifiers.
NASA EONET v3	REST search by date, category, or keyword	Wildfire, flood, storm, and general natural-event catalog records.
ReliefWeb	public update and report search	Humanitarian situation reports, response updates, and impact narratives.
HDX	CKAN <code>package_search</code> API	Candidate humanitarian datasets and country- or event-specific impact layers.
GDELT	event-window article search	Secondary news discovery used only when the package retains the result.
Wikipedia	REST page-summary and event-search interfaces	Event-name cross-checks and compact encyclopedic context.
Wikidata	entity-search API	Structured event and place identifiers used for identity checks.
NASA Earth Observatory	event articles and image pages	Locked event anchors and satellite-based physical narratives.
NASA Scientific Visualization Studio	visualization pages and media API	Official imagery, animations, and explanatory visual products.
NASA Disasters and Applied Sciences	activation and event pages	Satellite-based disaster-response context and mapped observations.
World Meteorological Organization	reports, topic pages, and bulletins	Authoritative global weather, climate, and extreme-event summaries.
World Meteorological Centre Beijing	event reports and operational bulletins	Regional meteorological summaries and event chronology.
NOAA Climate.gov	event trackers and climate reports	Reviewed US climate-event narratives and physical context.
US National Weather Service and WPC	forecast discussions, storm summaries, and precipitation pages	Operational weather chronology, rainfall totals, and warning context.
NOAA NESDIS and SSD	satellite product and event pages	Operational satellite imagery and environmental monitoring context.
Copernicus EMS	activation and rapid-mapping archives	Event confirmation and links to emergency-mapping products.
UN OCHA	situation reports and humanitarian updates	Reported impacts, needs, access constraints, and response activity.
UNOSAT	satellite-assessment reports and maps	Mapped damage, flood, and exposure evidence.
UN country teams and UN Geneva	country and event updates	Local consequence and response narratives retained as event evidence.
IFRC	emergency appeals and operational updates	Humanitarian consequences, needs, and Red Cross response records.
AHA Centre	regional disaster updates and reports	ASEAN event impacts and coordinated response context.

PACKAGE SOURCES Distinct source and access families across 199 event packages
continued

World Health Organization	health-emergency reports	Mortality, morbidity, service disruption, and public-health consequences.
UNICEF	humanitarian situation reports	Impacts on children, water, health, education, and relief delivery.
FAO	food, agriculture, and livelihood reports	Crop, livestock, food-security, and rural-impact evidence.
World Bank	assessment and recovery documents	Infrastructure, economic loss, and reconstruction context.
World Weather Attribution	event-study pages and reports	Attribution context and documented physical or societal drivers.
South Asian operational agencies	IMD/RSMC New Delhi, DHM Nepal, and DMC Sri Lanka	Official cyclone, rainfall, flood, and warning records.
National meteorological services	BoM, HKO, UK Met Office, MetService, and SAWS	National weather reports, warnings, and climate summaries.
Scientific institutes	ESSL, ICIMOD, NGI/NGU, ICL, OSTI	Hail, glacier, landslide, heat, and other specialist event accounts.

Meteorology, climate, hydrology, ocean, and air quality**13 families**

Provider or product	Access form	Role in a package
ERA5-Land	hourly reanalysis; package-local aggregates/stacks	Temperature, precipitation, wind, and land-surface background.
CHIRPS v2	daily precipitation archive	Event accumulation, wet spells, climatology, and rainfall anomalies.
NASA GPM IMERG V07	Earthdata CMR discovery and precipitation granules	Satellite precipitation totals, timing, and spatial structure.
Open-Meteo Archive	historical hourly/daily REST API	Point temperature, humidity, apparent temperature, rain, snow, and wind.
Open-Meteo Air Quality	historical air-quality REST API	PM _{2.5} , PM ₁₀ , aerosol, and related atmospheric fields.
NASA POWER	daily point REST API	Temperature, corrected precipitation, and 10-m wind cross-checks.
NOAA PSL climate indices	download tables for ONI, SOI, and related modes	Large-scale climate-state and teleconnection context.
NOAA CPC ONI	ASCII and tabular index archive	Operational Oceanic Niño Index values and episode timing.
IRI and Australian BoM	ENSO monitoring archives	Independent ENSO phase and coupled-ocean background.
GPCC	precipitation-anomaly products	Gauge-based anomaly context alongside CHIRPS.
NOAA OISST v2 high resolution	PSL THREDDS/ERDDAP catalog	Sea-surface temperature and marine-heatwave context.
Indian Ocean Dipole index	DMI archive	Indian Ocean climate-mode state for regional events.
NOAA ocean services	tide gauges and Ocean Prediction Center analyses	Coastal water levels, surge context, and marine storm structure.

Remote sensing and geospatial context**12 families**

Provider or product	Access form	Role in a package
NASA Worldview/GIBS	snapshot API and MODIS/VIIRS browse imagery	Pre-event and in-event true-color or thematic visual context.
MODIS/VIIRS vegetation products	Worldview or archive previews	Vegetation condition and drought-stress context.
MODIS/VIIRS MAIAC AOD	aerosol and smoke previews	Smoke transport and aerosol loading context.
Sentinel-1 GRD	VV pre/post imagery and change products	Flood, deformation, and surface-change comparison.
Sentinel-2 SR Harmonized	surface reflectance and dNBR derivatives	Burn severity and optical before/after consistency.

PACKAGE SOURCES Distinct source and access families across 199 event packages *continued*

Google Satellite Embedding / AlphaEarth	annual 64-band embeddings	Annual state vectors, cosine-change statistics, and change rasters.
Copernicus EMS Rapid Mapping	EMSR/GFM maps and archive products	Mapped flood, fire, earthquake, and exposure footprints.
Microsoft Planetary Computer	STAC metadata search	AOI/date discovery for registered Earth-observation assets.
Natural Earth	Admin-0 and physical basemap files	Low-resolution geopolitical and physical reference layers.
geoBoundaries	administrative-boundary API	Open administrative geometries for package AOIs.
GHSL	built-up and settlement layers	Urban extent and settlement structure.
Copernicus DEM and SRTM	elevation and derived slope	Terrain constraints for flood, landslide, and glacier analyses.

Exposure, infrastructure, and human impact

9 families

Provider or product	Access form	Role in a package
WorldPop GP 100 m	population grids and AOI summaries	Population exposure, density, and resampled 1-km context.
OpenStreetMap Nominatim	geocoding REST API	Event location, fallback coordinates, and AOI checks.
OpenStreetMap Overpass	bounded feature-query API	Roads, buildings, services, shelters, and critical amenities.
OpenStreetMap Geofabrik	regional extracts	Broader road and infrastructure context where bounded queries are insufficient.
FEWS NET and IPC	food-security products and reports	Drought exposure, food-security phases, and response pressure.
NYC Open Data	Socrata 311 service-request API	Flood complaints and city-service stress during urban rainfall.
Harris County and Houston open data	flood-warning and 311 interfaces	Gauge conditions, complaints, and local flood impacts.
Hong Kong public feeds	warnings, incident, and transport records	Local disruption and response evidence for typhoon events.
Brazil CEMADEN and civil defence	alerts and municipal reports	Rainfall-triggered hazard warnings and local consequences.

Hazard-specific scientific and operational sources

17 families

Provider or product	Access form	Role in a package
USGS ComCat	earthquake-event REST API	Origin time, magnitude, depth, location, and event products.
USGS ShakeMap and PAGER	event-product feeds	Instrumental intensity, shaking footprint, and loss context.
USGS Did You Feel It?	community-intensity product	Reported shaking distribution and populated-area impacts.
NOAA NCEI/WDS	Global Historical Tsunami Database	Tsunami source, run-up, and historical event records.
NOAA NHC	Tropical Cyclone Reports archive	Track, intensity, rainfall, surge, and impact summaries.
NOAA HRD H*Wind	surface-wind analysis archive	Storm wind structure and peak-wind calibration.
NOAA Coral Reef Watch	5-km thermal-stress products	HotSpot, Degree Heating Week, and bleaching-risk context.
Copernicus GloFAS/GFM/EFAS	flood-monitoring and event products	River-flood extent, timing, and European flood context.
Dartmouth Flood Observatory	Global Flood Monitor/archive	Independent flood occurrence and extent context.
Smithsonian GVP	volcano reports and event pages	Eruption chronology, volcano status, and physical interpretation.

PACKAGE SOURCES Distinct source and access families across 199 event packages <i>continued</i>		
Volcanic Ash Advisory Centres	operational ash advisories	Ash extent, altitude, movement, and aviation context.
GLIMS and Randolph Glacier Inventory	glacier outlines and catalog interfaces	Glacier geometry and inventory context.
NSIDC and ICIMOD	glacier and cryosphere reports	Regional ice change and glacier-hazard evidence.
NASA Global Landslide Catalog / COOLR	landslide catalog search	Occurrence records and landslide-event cross-checks.
USGS landslide releases	event pages, publications, and data releases	Site evidence for landslides and landslide-generated tsunamis.
ESSL and IEM/NWS archives	severe-weather summaries and Local Storm Reports	Hail, wind, tornado, and convective-event observations.
Australian fire and smoke products	official extent and smoke records	Fire perimeter, smoke, and air-quality context.
Package-local derivatives retained for computation		9 derived families
Provider or product	Access form	Role in a package
Compact event AOI	event-bounding-box derivative	Common spatial support for package layers.
ERA5-Land aggregate and stack	hourly statistics and raster bundle	Compact numerical access without replacing the upstream provenance.
GPM and CHIRPS event accumulations	derived totals and AOI rasters	Comparable precipitation ledgers across satellite and gauge products.
WorldPop summaries	AOI sum and 1-km resample	Deterministic population exposure inputs.
Sentinel-1 change	pre/post statistics and raster	Auditable radar change metrics.
Sentinel-2 dNBR	summary and raster	Auditable burn-severity calculation.
Satellite-embedding change	annual statistics and raster	Normalized annual surface-change evidence.
Bounded OSM slice	package-local JSON	Stable infrastructure and service counts.
Locked event anchor	package-local report copy	Fixed event identity, time window, and provenance for replay.

The inventory is exhaustive at this level of grouping, although each event uses only a hazard- and availability-specific subset. The 80 external rows cover distinguishable upstream sources and access methods; the nine derived rows record transformations reused across packages. Per-package manifests retain the raw label, exact URL, acquisition status, package-relative output, byte count, and checksum for record-level provenance.

C.4 SCORING, ISOLATION, AND AUDIT PROTOCOL

The judge receives the task, structured ground truth, final answer, reference process, rubric, and trajectory. It scores required answer units separately from evidence selection, calculation, source reconciliation, and mechanism analysis. Combined Core is the mean of these blocks, and Strict@95 records near-complete tasks. Capability columns use overlapping expert tags as diagnostic slices, not additive categories. Controlled comparisons use paired manifests and task-level bootstrap re-sampling.

Solvers may list, read, search, and compute only within the active event package. The reference solution, rubric, structured ground truth, and ground-truth program remain hidden. Tools record each path and operation before shortening observations, which preserves provenance through context compression. Finalization receives the accumulated research state and required schema but no answer-bearing hint. Released traces cannot reconstruct provider-hidden reasoning, proprietary routing, or undocumented cache accounting.

C.5 CONTROLLED-STUDY SAMPLING DESIGN

The reasoning-effort study uses a fixed manifest containing historically difficult tasks and a fixed-seed random draw from the rest of the release, preferring distinct events. Tool, routing, context, and interaction studies use a smaller paired manifest drawn from it. Oracle and robustness condi-

tions reuse the same task IDs and Top-20 router outputs. Paired 95% intervals use 5,000 bootstrap resamples of task-level score differences with seed 2026.

Professional-environment conditions use GPT-5.5 with the default medium reasoning setting and 48 turns. Core libraries retain the paired manifest; input-dependent environments draw a difficulty-balanced compatible manifest with seed 2026. Their compliance contract requires at least two successful, distinct specialist calls, targets three, and counts at most four. These paired ablations are diagnostic and should not be read as exact full-leaderboard effects.

Artifacts retained for audit.

- **Task identity.** Version, manifest, package checksum, and question reproduce the scientific input.
- **System identity.** Endpoint revision, controller, tool profile, and context policy reproduce the executable system.
- **Research trace.** Actions, arguments, observations, errors, and timing reconstruct the investigation, including failed calls.
- **Submission.** Raw text, parsed object, repair attempts, and final status keep malformed or missing answers in the denominator.
- **Reference and judgment.** Ground truth, solution, computation, rubric, judge prompt, and criterion decisions support answer-unit and aggregate recomputation.
- **Paired comparison.** Task IDs, intervention, seed, and bootstrap resamples identify the controlled change.

C.6 HUMAN EVALUATION AND AGENT-ASSISTED REVIEW

Human review begins when a task is conceived and continues through release. The team included six doctoral-level experts in Earth science and twenty undergraduates in related fields. Doctoral reviewers set the scientific direction, checked physical validity, and approved the final scoring specification. Undergraduate reviewers followed a common protocol for source and metadata checks, package replay, calculation verification, and secondary review of answer units. Reviewers contributed more than 50 hours per person on average.

Task direction and scoring checkpoints. For every task, a doctoral-level reviewer first analyzed the event and specified the scientific question, hazard process, spatial and temporal support, admissible evidence, required calculations, and answer fields. LLM agents then explored the package for plausible source combinations, alternative calculations, ambiguities, and likely failure modes. Reviewers used these passes to refine the reference analysis and define the final answer units and process checks. Agents proposed candidates; human experts decided which checks were scientifically necessary, supported by the package, and suitable for scoring.

Joint verification and correction. Every candidate task underwent agent replay followed by human inspection. Reviewers confirmed that package evidence could answer the public question, that `compute_gt.py` regenerated the structured targets, and that the solution, answer units, and rubric referred to the same quantities and physical claims. Any mismatch returned the task to construction. A small number of sources had minor spatial offsets between the data footprint and the documented disaster location. Human experts removed the affected answer units rather than score a quantity the package could not support.

Iteration scale. For difficult cases, constructing and reviewing a single question consumed more than 20 million tokens across repeated agent passes. These passes covered evidence exploration, draft construction, calculation checks, replay, discrepancy analysis, and revision. Human reviewers used them to align the scientific question, evidence, expert solution, deterministic program, answer units, and process rubric before release.

C.7 BENCHMARK CONSTRUCTION AND RELEASE GATES

Each construction stage produces an auditable artifact and an explicit release decision. A failed gate returns the task to the responsible stage.

Seven release gates.

1. **Event anchoring:** verify identity, hazard type, time window, spatial scale, and event–source agreement.
2. **Package assembly:** build a typed relative-path manifest and confirm that the required evidence roles are readable.
3. **Task authoring:** require package-local resolution without revealing a file path or answer-bearing value.
4. **Reference analysis:** align claims, units, mechanisms, and defensible alternatives; trace every load-bearing claim.
5. **Executable grounding:** regenerate structured ground truth deterministically and pass schema and tolerance checks.
6. **Rubric design:** map the task obligations to five to eight criteria totaling exactly 20 points.
7. **Final review:** align the question, solution, ground truth, program, and rubric before release approval.

D EXTENDED DIAGNOSTICS

Table 9 breaks Combined Core down by seven overlapping capability tags. Because a task may carry several tags, the columns are diagnostic views rather than additive partitions.

The strongest systems lead across nearly every tag, but their residual errors concentrate in the connections between results. Among the top five, causal-chain scores trail quantitative calculation by 4.9–7.9 points. A single defensible data path may be enough to compute a total or ratio. An evolving hazard account is harder to stabilize: rainfall, exposure, damage, and response evidence must refer to compatible windows and scales, and the proposed mechanism must survive their disagreement. GeoMMAgent breaks the otherwise stable ordering. Its short, structured search works well on tabular evidence but often bypasses raster intermediates, leaving remote sensing at 58.82. Capability scores thus capture the organization of an investigation as much as the knowledge of its backbone.

D.1 CAPABILITY-CONDITIONED PERFORMANCE

Table 9: **Core by capability.** Tags overlap; system groups follow Table 2. **Bold blue** marks column bests.

System	Earth-process analysis			Evidence synthesis			
	Physical mech.	Spatio-temporal	Quant. calc.	Multi-source	Causal chain	Ranking	RS / geo.
<i>Agent frameworks</i>							
GeoMMAgent	67.72	72.94	68.85	70.56	66.15	67.00	58.82
OpenResearcher	30.46	35.24	31.35	34.63	25.35	30.41	33.18
<i>Open-weight and Earth-specialized models</i>							
Qwen3-235B	31.90	40.35	34.03	35.51	29.00	29.29	32.32
GeohazardGPT	31.77	31.31	28.04	30.42	32.12	39.04	29.16
Qwen3-4B	24.63	22.71	24.70	24.32	25.89	24.06	32.61
GeoGPT	25.34	25.70	24.66	23.14	23.63	26.28	22.77
MiroThinker 8B	13.51	14.39	14.68	14.52	8.61	6.76	21.96
Nemotron Nano	11.79	14.41	12.30	17.89	12.42	13.26	15.48
Intern-S1-mini	12.04	10.70	11.99	10.45	10.57	13.21	16.50
<i>Hosted general-purpose systems</i>							
Claude Fable 5	85.15	85.51	87.14	83.45	82.20	80.98	89.99
GPT-5.6 Sol	81.81	83.31	85.42	82.13	77.54	71.63	88.50
GPT-5.6 Terra	80.83	79.83	82.96	79.06	76.26	76.12	86.27
GLM-5.2	80.25	77.30	80.24	76.85	80.08	78.34	79.27
GPT-5.5	77.21	78.38	81.91	76.79	74.33	75.76	83.77
Claude Sonnet 4.6	75.49	73.60	78.36	77.92	73.43	79.73	75.93
Kimi K2.5	75.07	79.10	77.81	75.34	70.48	70.82	77.94
DeepSeek V4 Flash	73.63	76.33	75.16	72.59	68.91	72.33	77.75
Tencent Hy3	72.67	75.92	73.71	70.32	69.72	70.23	79.41
Claude Haiku 4.5	69.60	72.44	76.09	73.50	67.73	66.76	79.66
Qwen 3.7 Plus	70.41	72.40	71.13	69.27	67.19	68.08	78.26
Doubao Seed 2.0 Mini	63.24	66.43	62.67	59.30	60.42	63.51	70.67
Gemini 3.1 Flash Lite	60.18	61.40	60.38	56.99	56.37	58.35	68.49
MiniMax M2.7	42.76	52.91	48.80	49.53	39.99	24.21	42.28
GPT-4o	39.97	38.86	38.77	34.93	37.56	43.70	48.30
GPT-5.6 Luna	34.74	42.91	38.58	36.48	31.77	21.17	38.21

D.2 PREFERENCE STRUCTURE AND DIAGNOSTIC ERROR PATHS

Pairwise judgments broadly recover the leaderboard order while exposing uncertainty hidden by Core alone. Figure 6 separates wins, losses, and ties in opponent-balanced comparisons. Leaders win consistently, adjacent frontier systems remain close, and ties are uncommon. The drop beyond the leading group agrees with low Strict@95: partial success is common, while consistently complete research remains rare.

Position. Reliability in dynamic Earth systems is the capacity to revise the evidence state. As new observations arrive, a system must detect conflicts, isolate local errors, and carry corrections into derived quantities, causal interpretations, and the final event account. Scientific agents differ not only in how often they err, but in whether errors are found, contained, and repaired; longer traces, more tool calls, or higher execution success cannot substitute for that capacity (Chen et al., 2025; Du et al., 2025).

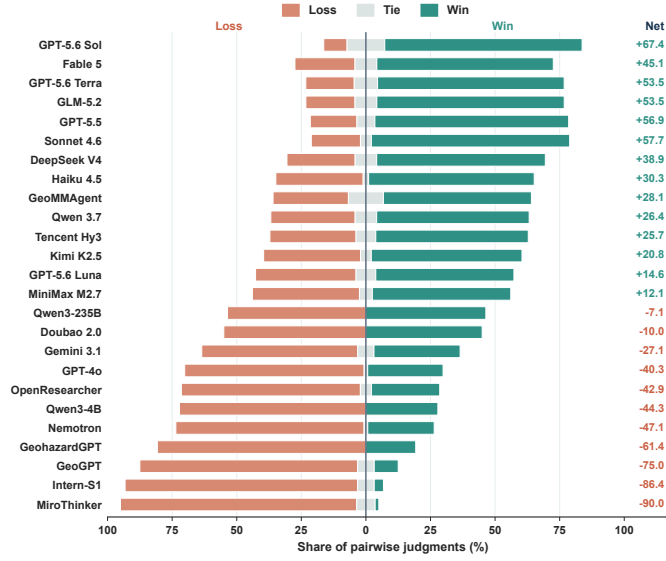


Figure 6: Pairwise win, tie, and loss shares. Net is win share minus loss share.

Figure 7 shows the same separation at task level. Frontier models have a dense high-score mode but retain a long tail toward zero. Agent frameworks cluster at moderately high scores, while open-weight and Earth-specialized systems shift downward and rarely approach complete solutions. Error counts overlap far more than Core scores. Strong systems still make mistakes; their advantage lies in catching or containing them before the final answer. Error frequency therefore describes the re-search process, while Core records the damage that remains.

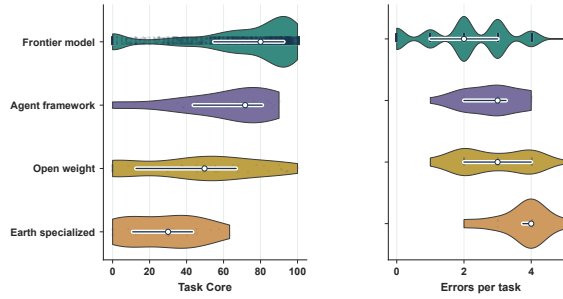


Figure 7: Task Core and diagnosed errors by system family.

The family-level flow in Figure 8 sharpens this result. Computation and answer errors occur in every family, yet their consequences differ. Frontier systems more often keep affected tasks in the 40–70 or above-70 bands; open-weight and Earth-specialized systems more often fall below 40. Recovery, cross-source checking, and final synthesis determine whether a local mistake remains local or spreads through the answer.

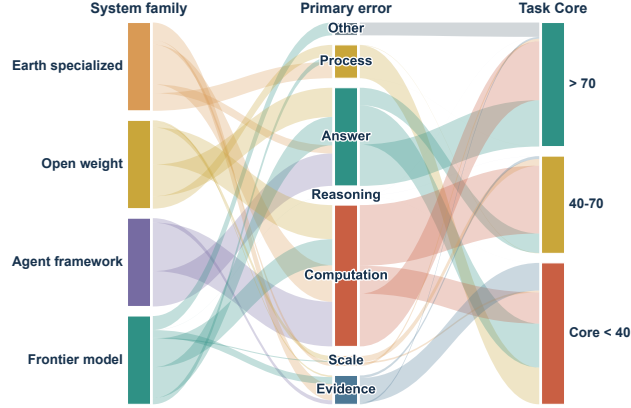


Figure 8: System family, primary error family, and task-level Core band.

Figure 9 examines the same pattern within systems. The largest Core losses accompany premature stopping, failed recovery, tool failure, and missing required outputs—failures that interrupt the research loop itself. Numerical and schema errors are common, but their average effect is smaller because some runs catch or absorb them before finalization. The inset connects missing evidence, output omissions, schema mistakes, and calculation errors into one cluster. These are often successive stages of a failed investigation, not independent defects.

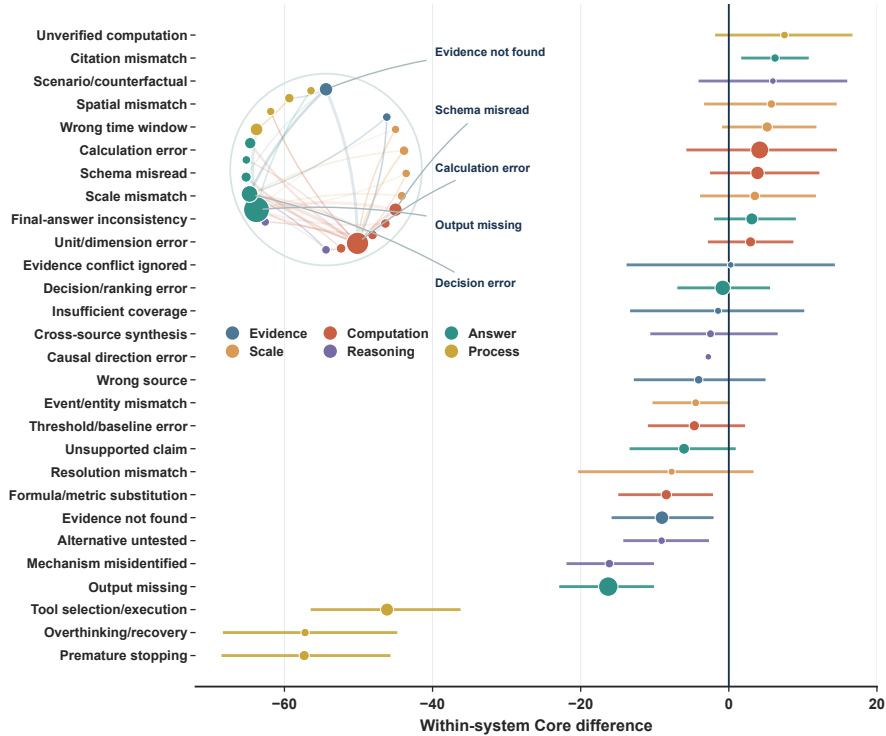


Figure 9: Within-system Core differences and the strongest error co-occurrences.

D.3 REASONING MODES AND RESOURCE ACCOUNTING

Tables 10–12 report trajectory, latency, token use, and evaluation cost. They separate interactive retrieval from direct serialization and model activity from successful evidence acquisition.

D.3.1 INTERACTIVE TRAJECTORY DIAGNOSTICS

Higher reasoning effort changes the treatment of evidence more than the amount collected. At xhigh, the EarthVerse harness opens about as many files as at medium effort and uses fewer rounds, yet takes almost twice as long. The additional work occurs between observations: the model compares interpretations, revisits calculations, and decides whether its account is complete. Explicit review intensifies this pattern. From harness-high to review-xhigh, output nearly doubles while file reads barely move. Some of that deliberation repairs omissions; some only restates a settled path. Rounds, tokens, and latency therefore measure different kinds of work, and none is a reliable stand-in for scientific progress.

Table 10: **Interactive trajectory diagnostics.** Per-task means; call success is the share of tool calls that return normally.

Mode	Effort	Execution			Evidence			Resources				
		Rounds	Tool calls	Success (%)	Reads	Files	Evidence refs.	Input (k)	Output (k)	Total (k)	Cached (k)	Latency (s)
EarthVerse	none	9.68	8.68	92.08	3.84	7.00	9.94	94.3	2.6	96.9	10.6	140
	low	14.70	13.70	93.67	5.64	9.44	14.70	157.4	5.1	162.4	15.4	259
	medium	17.20	16.20	95.39	6.84	10.71	18.12	198.6	7.9	206.4	17.9	424
	high	14.96	13.96	97.19	5.15	9.92	17.19	164.8	9.5	174.3	13.2	571
	xhigh	11.25	10.25	100.00	4.75	10.25	16.50	90.6	9.4	100.0	15.9	674
Review	none	11.62	9.62	94.26	4.96	7.42	10.46	126.9	3.8	130.7	18.8	108
	low	18.18	16.18	94.94	6.84	10.60	17.02	219.0	8.0	227.0	25.1	253
	medium	20.81	18.81	94.78	6.56	11.88	21.02	253.8	12.5	266.3	25.2	552
	high	20.63	18.63	95.37	6.50	12.13	21.10	239.2	15.9	255.1	22.2	1,105
	xhigh	17.57	15.57	99.29	5.86	10.14	18.29	198.9	18.8	217.7	25.6	1,493

D.3.2 DIRECT PACKAGE-SERIALIZATION DIAGNOSTICS

Direct conditions receive a fixed package serialization before generation. Compressed direct includes more files in far fewer tokens than all-text, yet neither mode lets the model return to a source after discovering that a window, unit, or variable was wrong. That restriction changes the task. The model can reason about the supplied snapshot, but it cannot conduct the next measurement implied by its own reasoning. Larger inputs and longer outputs fail to rescue these baselines because the evidence schedule is fixed before the analysis begins. The missing operation is revision of the observation set, not another pass over the same text.

Table 11: **Direct-serialization resources.** Direct modes inject a fixed file set before generation; values are per-task means.

Mode	Effort	Injected files	Evidence refs.	Input	Output	Total	Latency (s)
Compressed	none	17.92	24.56	13,016	878	13,894	55.61
	low	17.92	25.31	13,033	1,857	14,890	120.45
	medium	17.92	25.72	12,838	3,004	15,842	152.36
	high	17.93	26.73	12,958	3,630	16,588	71.47
	xhigh	17.67	25.00	12,878	4,726	17,604	88.14
All text	none	7.57	13.89	80,842	751	81,593	32.06
	low	7.68	14.00	80,886	1,854	82,739	44.10
	medium	7.46	13.90	80,941	2,557	83,498	61.02
	high	8.73	15.13	81,623	3,735	85,359	78.49
	xhigh	8.00	10.00	83,655	4,013	87,668	94.38

D.3.3 EVALUATION COST

Table 12 reports the cost of the evaluated runs for models with a comparable USD list price.

Table 12: **Evaluation cost.** Rows descend by total cost.

System	Core	Tokens/task (k)	Price (USD/M)	Cost (USD)
Claude Fable 5	84.97	161.6	50.00	3,271.57
GPT-5.5	78.06	224.9	30.00	2,732.09
GPT-5.6 Sol	81.68	205.9	30.00	2,501.58
Claude Sonnet 4.6	76.16	331.0	15.00	2,011.07
GPT-5.6 Terra	80.06	182.6	15.00	1,109.03
Claude Haiku 4.5	72.23	477.0	5.00	966.02
GPT-4o	39.23	127.4	10.00	515.92
GPT-5.6 Luna	35.66	156.9	6.00	381.31
DeepSeek V4 Flash	73.54	389.7	0.28	44.19
Gemini 3.1 Flash Lite	59.66	70.1	1.50	42.61

Quality and cost do not form a simple ladder. The strongest frontier evaluations exceed \$2,500, while GPT-5.6 Terra retains about 98% of GPT-5.6 Sol’s Core at 44% of its cost. The two Flash systems remain below \$45 but differ by almost 14 Core points. A long trace from a low-price model may still cost less than a short frontier run, so quality and cost should be reported together when planning replication.

D.4 TOOL, MEMORY, AND INTERACTION ABLATIONS

Table 7 reports outcome and capability scores for each ablation; Table 13 shows how the same runs searched, computed, and consumed context. Token use is descriptive and does not affect scoring.

D.4.1 RESEARCH BEHAVIOR AND CONTEXT CONSUMPTION

The trajectory diagnostics expose two different kinds of false productivity. Under aggressive compression, the agent lists files more than four times as often as with long retention and records more evidence references, yet receives the lowest process score. It is rebuilding context that the controller discarded, not broadening the analysis. The expert-only suite looks efficient for the opposite reason: it has the highest call success while inspecting the fewest distinct files. Its tools run cleanly, but the investigation is narrow. Activity counts become meaningful only when tied to state changes: a new source should resolve a claim, a computation should test a mechanism, and a repeated read should have a reason.

D.4.2 CAPABILITY-CONDITIONED EFFECTS

The largest capability losses occur when evidence must remain connected across several steps. With 170 visible tools, multi-source performance falls to 47.50 even though most other catalog configurations stay near or above 88. Aggressive compression loses 20.07 points in causal-chain reasoning relative to long retention. The two interventions damage continuity in different places. A large catalog makes the next action unstable; compression makes the meaning of earlier actions unstable. A single calculation can survive either disturbance because its inputs are local. A physical account cannot: its validity depends on remembering why several observations belong together.

Table 13: **System-ablation trajectories.** Per-task means for the panels in Table 7.

Condition	Tools			Evidence		Actions			Process	Tokens			
	Calls OK	Calls failed	Success (%)	Files	Evidence refs.	Lists	Searches	Python	Process (%)	Input	Output	Total	Cached
<i>(a) Tool-catalog scale</i>													
Fixed tools (6)	15.80	2.15	92.52	12.75	19.80	1.95	4.45	5.45	65.62	219,639	9,918	229,557	25,882
Fixed tools (10)	15.25	1.75	92.22	11.05	17.40	1.30	3.30	3.35	57.10	185,888	8,159	194,047	25,408
Fixed tools (20)	15.55	2.05	91.74	12.05	17.15	1.45	3.10	4.60	59.38	223,666	9,246	232,912	25,267
Fixed tools (40)	15.75	1.95	92.81	11.95	19.70	1.55	4.15	4.10	61.00	224,175	8,818	232,993	24,346
Fixed tools (70)	15.00	1.90	92.39	11.40	18.35	1.25	3.35	2.85	61.25	224,853	8,906	233,759	26,445
Fixed tools (100)	16.35	2.00	92.54	12.00	20.25	1.10	3.90	3.60	63.12	268,527	10,409	278,936	32,294
Full tool catalog (170)	13.90	1.50	95.04	9.55	13.15	1.05	1.85	2.55	55.88	264,983	8,233	273,216	33,331
<i>(b) Semantic routing and tool composition</i>													
Top-10 dynamic router	14.65	2.15	93.16	12.40	18.65	1.40	3.50	4.70	66.25	205,908	10,319	216,227	29,286
Top-20 dynamic router	12.75	1.65	92.98	11.30	16.15	1.35	2.60	4.05	65.88	170,972	9,038	180,010	21,696
Workflow-aware router	15.20	1.95	93.38	12.25	21.65	1.40	4.65	3.45	56.75	210,928	9,304	220,232	26,381
Expert-only suite	17.70	0.60	97.59	7.35	12.60	0.00	0.00	0.00	60.88	283,502	11,988	295,490	24,192
General + expert suite	14.05	1.55	94.43	10.40	15.40	1.40	3.05	3.45	63.00	186,216	8,875	195,092	31,437
Atmospheric Python suite	14.60	1.35	94.78	10.35	18.40	1.45	4.10	2.85	60.12	191,208	7,593	198,802	25,344
Atmospheric Fortran suite	14.55	1.85	93.68	11.25	17.10	1.55	3.35	3.30	57.88	195,439	9,823	205,262	31,078
<i>(c) Context retention</i>													
Aggressive compression	24.60	1.95	93.88	12.50	34.90	5.40	6.25	4.00	51.62	187,492	9,545	197,038	10,765
Short retained context	20.20	2.25	93.64	13.75	26.80	2.70	6.40	5.10	56.38	198,062	11,765	209,827	14,682
Long retained context	13.35	1.85	92.77	11.80	16.95	1.20	3.70	4.45	63.38	218,347	9,189	227,536	48,998
Full retained context	12.15	1.30	93.84	11.50	18.00	1.05	3.90	3.40	58.25	180,517	7,182	187,700	80,819
<i>(d) Interaction budget</i>													
12-turn budget	9.95	0.30	97.36	8.50	13.05	1.15	2.70	1.30	49.50	101,826	4,437	106,263	23,002
24-turn budget	12.90	1.65	93.11	10.80	16.90	1.35	4.20	3.30	52.88	153,261	8,143	161,404	32,346
96-turn budget	13.60	1.30	94.12	11.90	18.45	1.60	3.85	4.55	60.12	173,925	8,873	182,798	27,763
Unbounded turn budget	15.20	1.80	94.22	12.30	21.40	1.60	5.15	4.35	64.50	211,751	9,429	221,180	30,131

D.5 ORACLE, INTERFACE, AND EVIDENCE ROBUSTNESS

Table 4 consolidates outcome scores, paired uncertainty, and capability slices for each intervention. Table 14 separately reports trajectory and resource effects. Confidence intervals resample paired task-level Core differences. Oracles provide guidance without answers, while robustness conditions alter only package evidence.

D.5.1 PAIRED INTERVENTION UNCERTAINTY

Only the two evidence-localization oracles have intervals that exclude zero: relevant files [+5.48, +24.69] and the evidence map [+2.34, +21.25]. The stable gain comes from knowing where evidence is, not from receiving a generic plan. A plan names familiar analytical moves; localization resolves an event-specific uncertainty that scientific knowledge alone cannot settle. The other interventions remain suggestive because their intervals cross zero. On a diagnostic manifest, a few packages whose structure favors one interface can still move the mean substantially.

D.5.2 TRAJECTORY AND RESOURCE EFFECTS

Oracle guidance shortens search without necessarily reducing cost. The evidence map lowers rounds and file reads, but the injected structure itself consumes context and takes time to interpret. Distractors produce the reverse pattern: the agent performs more actions and records more evidence references with almost no quality loss. Obvious clutter is therefore mostly an efficiency tax. A compact evidence description poses a different problem. The agent must decide whether to trust the summary or reopen the underlying source, and the latter is often necessary to recover metadata, aggregation choices, or conflicts. Search length and evidential control are related, but they are not interchangeable.

Table 14: **Intervention trajectories.** Panels match Table 4; values are per-task means.

Intervention	Execution			Evidence and compute			Latency		Tokens			
	Rounds	Tool calls	Success (%)	Reads	Files	Evidence refs.	Python	Seconds	Input	Output	Total	Cached
<i>(a) Answer-free information oracles</i>												
Relevant-file oracle	15.30	14.30	92.17	5.15	11.30	16.10	4.15	217.26	181,939	5,572	187,512	19,507
Evidence-map oracle	11.10	10.10	93.28	3.10	8.55	11.10	2.75	436.56	290,368	5,050	295,418	62,950
Tool-subset oracle	17.80	16.80	93.18	6.60	13.35	21.60	3.90	212.74	205,252	5,628	210,880	27,738
Planning oracle	16.95	15.95	91.49	5.45	12.65	18.30	4.05	185.18	196,489	5,817	202,306	25,190
<i>(b) Tool-interface presentation</i>												
Hierarchical tool catalog	16.60	15.60	91.97	5.75	12.45	17.75	3.85	197.77	182,350	6,000	188,351	20,416
Descriptive tool names	18.25	17.25	93.42	5.40	12.25	19.65	3.65	150.14	202,390	5,462	207,853	28,749
Dynamic tool descriptions	17.55	16.55	92.12	4.00	12.85	19.95	3.65	182.42	199,673	5,036	204,709	23,053
Opaque tool names	17.90	16.90	92.44	7.00	10.85	16.50	4.30	234.10	215,650	6,294	221,943	23,027
<i>(c) Evidence perturbations</i>												
Equivalent unit change	16.45	15.45	90.70	5.20	13.25	19.25	3.90	176.16	184,679	5,153	189,832	19,904
Added distractor files	19.60	18.60	89.97	6.20	13.10	22.50	4.15	162.54	220,098	5,658	225,756	25,792
Missing evidence	17.40	16.40	89.12	5.65	12.05	21.20	3.55	181.47	198,173	5,219	203,392	19,546
Truncated evidence	16.10	15.10	91.65	4.25	12.75	22.10	3.55	265.79	184,244	5,827	190,071	16,781
Conflicting evidence	16.50	15.50	92.04	5.65	12.55	19.70	3.25	155.59	182,815	5,875	188,689	18,624
<i>(d) Optional meteorological access</i>												
Compact meteorology suite	17.95	16.95	94.90	5.85	12.75	20.60	4.05	223.20	201,057	5,822	206,879	27,571
Cue-routed meteorology suite	15.65	14.65	92.50	5.65	11.50	17.85	3.15	238.22	170,431	5,875	176,306	25,882

D.5.3 CAPABILITY-CONDITIONED INTERVENTION EFFECTS

The effective oracles help most with spatiotemporal reconstruction: both exceed 90, compared with 64.40 at baseline. File localization directly removes uncertainty about which records contain the event window and region. Ranking and decision improve less because the oracle does not say how competing quantities should be weighted. Opaque tool names damage the same capability most strongly. Reconstructing a window rarely depends on one call; it requires a sequence of filtering, reading, aggregation, and comparison. Losing the semantic role of any operation makes the resulting timeline harder to recover and harder to audit.

D.6 PROFESSIONAL METEOROLOGICAL ENVIRONMENTS

The professional-environment study separates method availability from execution. Compliance records whether the required interactions occurred; call success records whether they completed without adapter, argument, or runtime failure.

D.6.1 COMPLIANCE, EXECUTION RELIABILITY, AND IMPLEMENTED SCOPE

Nearly every run attempts the required specialist interaction, so willingness to call the tool is not the bottleneck. Execution is. Specialist-call success ranges from 20.29% for PyET to 97.56% for R climate, and high-success environments usually need fewer repair rounds and smaller contexts. The ranking should not be read as a comparison of languages. It reflects the overlap between the audited operations and the data actually present in the packages. R climate, Julia, and CDO often accept the available time series or grids directly; other adapters require variables, record lengths, or sampling structures that many event packages do not contain.

Table 15: **Scientific-environment outcomes and execution.** Final Answer, Process, and Core scores are reported with per-task compliance, successful specialist calls, resource use, and implemented operations.

Environment	Operations	Outcome			Execution		Resources		
		Answer	Process	Core	Comply (%)	Success (%)	Rounds	Tokens	Latency (s)
<i>(a) Python scientific libraries</i>									
MetPy	rain summary; profiles; kinematics	68.69	70.38	69.53	100	38.83	25.00	311,043	255.60
xclim	heat/rain extremes; wet/dry spells	67.77	69.62	68.70	95	38.78	24.50	299,074	268.44
climate-indices	SPI; SPEI; PET	64.50	66.75	65.62	95	22.60	30.55	374,199	306.69
PyET	Penman–Monteith; Hargreaves	65.34	67.38	66.36	95	20.29	32.30	387,917	315.96
thermofeel	UTCI; heat index; wind chill	66.78	68.75	67.76	100	26.85	28.55	346,924	357.88
pyextremes	block maxima; POT; return levels	59.02	60.00	59.51	100	34.19	28.05	363,839	289.16
xskillscore	RMSE; correlation; Brier; CRPS	63.37	64.12	63.75	95	57.14	23.70	297,724	231.47
<i>(b) Compiled and domain-language environments</i>									
Julia	summary; integration; sensitivity	71.77	70.25	71.01	100	91.11	21.50	256,852	204.47
R climate	summary; SPEI; GEV; trend	68.55	71.38	69.96	100	97.56	21.70	259,984	190.20
CDO	temporal and field aggregates	66.91	69.50	68.20	100	81.63	22.30	268,906	197.90
NCO	inspect; weighted mean; derive field	67.39	65.00	66.19	100	35.59	26.65	321,773	230.95
NCL	summary only	65.03	65.00	65.01	95	40.62	25.70	308,559	270.94
Fortran	rain; kinematics; moisture transport	64.46	64.25	64.35	95	43.48	25.70	301,236	222.60

Applicability and reproducibility. Environment names refer to the audited adapters in Table 15, not to every feature of the upstream packages. The outcome columns make the distinction explicit: R climate has the highest call success, while Julia has the highest Core. A successful call proves only that code ran. It does not establish that the estimator fits the record, that the sampling geometry supports the comparison, or that the returned value answers the question. Those checks sit between execution and scientific use, and the gap between execution reliability and final quality shows why they cannot be collapsed into one metric.

Each adapter therefore records normalized arguments, status, output, and errors. It also declares the operations it implements and validates package-scoped inputs before execution. This makes a justified failure reproducible: another reviewer can see that a method was rejected because the record was too short or a required variable was absent. Merely rerunning a command is weaker. Scientific reproducibility requires reconstructing why the operation was admissible and how its output entered the claim.

E PROMPTS AND EXECUTION CONTRACTS

The prompts are grouped by role: package-scoped solving and review, answer–process evaluation, and benchmark construction. Placeholders mark values supplied by the harness. The solver sees no demonstration, hidden reference, or recommended file. Appendix G presents representative runs.

Runtime visibility. The solver sees the public question, event-package root, exposed tool schemas, and package-relative observations. Ground truth, reference solutions, rubrics, computation programs, judge records, and other model answers remain hidden. The trace retains both successful and failed tool calls. Depending on the condition, the controller may request two format repairs, run a bounded evidence review, require a specialist operation, or reserve the final call for a schema-valid submission. None of these transitions adds scientific evidence. Appendix C.2 records their limits.

PROMPT	RUNTIME	Package-scoped research suite
SOLVER EarthVerse interactive solver		
SYSTEM You are an autonomous extreme-event benchmark solver. You must solve the task by planning, selecting local files, executing tools, reading observations, and then producing a structured final answer. Use only local event-package evidence returned by tools. Do not use web knowledge. Never read computed_gt.json,		

PROMPT

RUNTIME

Package-scoped research suite

continued

solution_en.md, review.json, or compute_gt.py. Prefer deterministic Python calculations when the task asks for metrics. You must make real tool calls before finalizing. Every assistant message must follow one of the two response contracts below.

TASK MESSAGE

Task id: <<TASK_ID>>
 Event id: <<EVENT_ID>>
 Question path: <<QUESTION_PATH>>
 Event package path: <<PACKAGE_ROOT>>

Question:
 <<PUBLIC_QUESTION>>

Evidence boundary:
 Strict package-only mode: only the current question and event package evidence are visible.

Tools:
 <<REGISTERED_TOOL_CATALOGUE>>

TOOL ACTION CONTRACT

Action	tool_call
Tool	one registered tool name
Arguments	only the arguments required by that tool
Reason	why this call is needed for the current analysis

FINAL ANSWER CONTRACT

Action	final_answer
Answer	the exact structure requested by the public question
Confidence	one numeric confidence value
Evidence	for each claim: package-relative source file and support type (direct metric, report claim, metadata, image context, table value, derived metric, or unsupported boundary)
Research	a brief ordered summary of the reasoning
Limitations	important unsupported or unresolved elements

Start by inspecting the package inventory, then choose only the files needed for this event-specific disaster analysis.

REVIEW Final evidence review

Begin a final evidence-review phase before submitting the answer. Treat the prior answer as a draft: re-check its evidence provenance, formulas, units, intermediate values, competing physical explanations, and every required output field against the package. Use additional tools when verification is useful. Return either one tool action or a revised final answer; do not merely endorse the draft. You have at most <<MAX_REVIEW_TURNS>> review turns.

PROMPT

EVALUATION

Joint answer and process judge

JUDGE Answer and research-process evaluation**SYSTEM**

You are a strict but fair evaluator for a local extreme-event benchmark. Your job is to judge answer correctness and task-rubric quality. Use partial 0-100 scores and avoid old deterministic exact matching.

CALIBRATION CORE

Evaluate this model response for an extreme-event evidence-and-reasoning benchmark. Make one combined evaluation and produce two core 0-100 scores.

- answer_correctness_score is the direct partial-credit comparison between the model's final answer and the ground truth. Treat the answer as a set of units: required fields, labels, decisions, values, units, orderings, alternatives, stage boundaries, and required conclusions. Missing required fields are

PROMPT

EVALUATION

Joint answer and process judge

continued

incorrect; correct prose cannot replace a missing structured field.

- llm_rubric_score must come from the task's own 20-point rubric. Compare the answer and visible trace against every criterion, assign earned points out of 20, and convert the result to percent.

- Do not invent extra credit. A plausible narrative, many tool calls, or reading the right files is not by itself a correct solution process. Evidence, intermediate values, transformations, and conclusions must agree.

- Keep the two scores distinct. A correct-looking final answer cannot substitute for missing process evidence, and an attempted method cannot receive full rubric credit when its result is wrong.

- Use only the supplied package evidence, ground truth, task solution, rubric, final answer, and model trace. Do not rely on web knowledge.

Runtime payload:

<<TASK_METADATA>>

<<QUESTION>>

<<GROUND_TRUTH>>

<<SOLUTION_AND_20_POINT_RUBRIC>>

<<MODEL_FINAL_ANSWER>>

<<MODEL_TOOL_TRACE>>

<<MODEL_RUN_NOTES>>

EVALUATION RECORD

Scores answer correctness and rubric-based process quality

Answer units each required unit and whether it was satisfied

Rubric ledger criterion-level points earned out of 20

Diagnosis verdict, critical errors, missing elements, unsupported claims

Rationale one sentence explaining the score pair

PROMPT	CONSTRUCTION	Benchmark construction suite
		QUALITY Task construction acceptance review ROLE You are the final quality reviewer for one extreme-event benchmark task. INPUTS - The public question (question_en.md) - The expert solution (solution_en.md) - The executable ground-truth program (compute_gt.py) - The generated scoring contract (computed_gt.json) - The optional review record (review.json) TASK Verify that the five artifacts describe one event-specific scientific target. Do not approve a rewritten public question when the solution or GT still scores an older package-inspection objective. ACCEPTANCE CHECKS 1. The public question reads like a real disaster-analysis assignment. 2. It contains no local path, visible CSX identifier, package inventory, available-material wording, AOI, quality-gate, evidence-boundary, or benchmark phrasing, and it does not reveal the useful files. 3. The solution contains Final Answer, Key Computations, Reasoning Path, Computed Interpretation, and Scoring Rubric. 4. computed_gt.json declares schema_version 1.0, task and event identifiers, task type, evaluation mode, primary GT, key values, and a top-level rubric. 5. Deterministic tasks contain compact machine-checkable targets. Open expert tasks contain concrete judge anchors rather than a fabricated exact essay. 6. The rubric has 5-8 criteria whose points sum exactly to 20. 7. Former distractors are rejected through numeric tests, inequalities, process timing, or competing-mechanism checks rather than option prose. 8. compute_gt.py runs successfully and regenerates the scoring contract. OUTPUT Return changed or reviewed file paths, event and hazard type, task type, evaluation mode, acceptance status for every check, and any remaining ambiguity, missing-data risk, temporal/spatial mismatch, or unsupported overclaim.

PROMPT**CONSTRUCTION****Benchmark construction suite***continued***RUBRIC** Event-specific rubric construction**ROLE**

You are constructing the hidden scoring contract for one benchmark task.

TASK

Create a 20-point rubric that can be applied without guessing the author's intent. Use 5-8 small, independently judgeable criteria. Every criterion must contain name, points, description, and partial_credit. The point values must sum exactly to 20.

REQUIRED COVERAGE

- 3-4 points: final conclusion, numeric target, threshold state, proof result, ranking, or concise canonical stance.
- 3-5 points: required quantitative extraction or calculation, including units, tolerances, and windows where relevant.
- 2-4 points: formula, index, process-window, causal-chain, or score-ledger logic.
- 2-4 points: fusion of reports, time series, rasters, imagery, exposure, infrastructure, or trajectory evidence into checkable anchors.
- 1-3 points: a compact event-specific consequence that follows from the computed evidence.
- 1-3 points: rejection of plausible competing mechanisms or overclaims.
- 1-2 points: complete, concise output in the requested structure.

HARD RULES

Do not use vague criteria such as "good reasoning" or "uses evidence well." Do not award points for choosing an option after a converted task has become a numeric or structured analysis. A criterion that depends on an incorrect scientific result cannot receive full process credit. For llm_judge tasks, include canonical_stance, required_claims, required_metrics, forbidden_claims, and acceptable_variants in primary_gt. For hybrid tasks, preserve both the deterministic core and the reasoning anchors.

RUBRIC RECORD

Total points	exactly 20
Criteria	5-8 independently judgeable items
Each item	event-specific name; points; full-credit requirement; partial-credit and error rule

SOURCE Event authenticity and source audit**ROLE**

You are auditing the real-event anchor of one event package.

QUESTIONS

1. Did the event occur? Require at least one traceable source confirming the event name, time window, or principal affected region.
2. Is the source reliable? Prefer government agencies, international organizations, peer-reviewed work, and authoritative scientific institutions; then disaster catalogues and major news. Encyclopedias and generic search pages are supporting evidence only.
3. Does the source actually correspond to this event? Check event name, time, place, and hazard type. A source about the same hazard family but a different time or place is not a strong match.

OUTPUT FIELDS

- existence_verdict: confirmed | likely | weak | not_confirmed
- temporal_match: exact | near | broad | mismatch | unknown
- spatial_match: exact | regional | broad | mismatch | unknown
- hazard_match: exact | compound_related | broad_family | mismatch | unknown
- source_reliability: high | medium | low | mixed
- release_decision: keep | keep_with_minor_fix | needs_stronger_source | replace_event
- concise evidence and repair notes

DECISION RULES

Use keep when the event is confirmed, at least two of temporal/spatial/hazard

PROMPT	CONSTRUCTION	Benchmark construction suite	<i>continued</i>
matching are exact or regional, and the preferred evidence is authoritative. Use <code>keep_with_minor_fix</code> when the event is real but URL, naming, or scale needs a small correction. Use <code>needs_stronger_source</code> when current support is generic, secondary, or overly broad. Use <code>replace_event</code> when the event cannot be confirmed or the cited evidence clearly belongs to another event.			

F REPRESENTATIVE CASE SET

The six cards below index remote-sensing reconciliation, urban pluvial causality, evidence substitution, multi-agent quantitative decomposition, integrated heat research, and a partial scientific failure. Together they cover five system families without repeating the full records that follow.

REPRESENTATIVE CASE SET			
■	FRONTIER HOSTED	GPT-5.6 Sol	CSX-189_Q1
Core 99.50 Uljin wildfire consistency. Report claims, wind and precipitation, Sentinel-2 dNBR, annual change, smoke transport, and bounded impact claims. TRACE Sixteen successful calls, including a failed semantic search recovered by a filename query and a final provenance check.			
■	GENERAL HOSTED	Tencent Hy3	CSX-059_Q2
Core 100.00 NYC Ida pluvial overload. An hourly rainfall record inside a two-day rain shield, population load, surface change, and a controlled future perturbation. TRACE Ten evidence calls followed by one deterministic calculation of every ratio, normalization term, and scenario score.			
■	OPEN-WEIGHT LARGE	Qwen3-235B	CSX-006_Q2
Core 93.93 Heat-source substitution audit. Variable-level admissibility of an ERA5-Land aggregate substituted for local hourly apparent-heat and warm-night data. TRACE Eight calls covering inventory, source comparison, schema checks, corrected heat metrics, and numerical source impact.			
■	MULTI-AGENT FRAMEWORK	GeoMMAgent	CSX-063_Q1
Core 83.50 Zhengzhou threshold ledger. Reported burst and multiday rainfall, product ordering, exposure thresholds, and a conjunctive flood diagnosis. TRACE Four coordinator roles and nine package-tool calls for retrieval, calculation, scientific synthesis, and schema assembly.			
■	RESEARCH FRAMEWORK	OpenResearcher	CSX-018_Q2
Core 86.92 Sahel heat-response model. Hourly heat, wet bulb, warm nights, exposure, services, roads, surface stability, two indices, and a +1 C counterfactual. TRACE Twenty-one calls, including one failed heat script, an explicit repair, and independent checks of peaks, duration, density, scores, and deltas.			
■	GENERAL HOSTED	GPT-4o	CSX-312_Q2
Core 41.25 Partial scientific failure. The model identifies the coupled Kilauea mechanism but reports lava effusion rate instead of the required summit-collapse-to-flow volume ratio. TRACE Nineteen model rounds and nine tool calls: seven succeed, two guessed paths fail, and the final answer earns Answer 40.00 and Rubric 42.50.			

G DETAILED RESEARCH TRAJECTORIES

Appendix F identifies the six selected systems and tasks. This section expands each card into its scientific target, evidence sequence, calculations, and decision. The Uljin case retains the complete task and action order; the remaining records focus on the steps that determine the result.

G.1 FULL CASE: ULJIN REPORT-TO-BURN CONSISTENCY

QUESTION CSX-189_Q1**SCIENTIFIC TARGET****Uljin wildfire report-to-burn consistency ledger**

A technical review team is checking whether the March 4–13, 2022 Uljin wildfire is numerically consistent with a dry, wind-assisted burn–scar diagnosis rather than a late–rain–control or image–baseline alternative.

EVIDENCE SCOPE

Use only package–local evidence. Do not use point–weather products for this task; build the ledger from the event report, package gridded precipitation and wind summaries, Sentinel–2 dNBR, and annual embedding–change context.

REQUIRED COMPUTATION

Compute these ledger values:

- report_dry_wind_flag: 1 if the report links the event to strong winds and dry weather, otherwise 0.
- smoke_transport_flag: 1 if the report states that smoke moved toward southern Japan, otherwise 0.
- era5_mean_wind_speed_mps: vector speed from mean 10 m u and v components.
- era5_wind_bearing_to_deg: direction toward which the mean wind vector points, degrees clockwise from north.
- precip_mean_spread_mm: spread between ERA5–Land, GPM IMERG, and CHIRPS event precipitation means.
- dnbr_mean: mean dNBR over the event burn–change window.
- dnbr_max_to_mean_ratio: dNBR maximum divided by dNBR mean.
- dnbr_to_annual_change_mean_ratio: dNBR mean divided by annual 1–minus–cosine embedding–change mean.
- reported_charred_area_km2: reported hectares converted to square kilometers.

MECHANISM CHECKS Apply these tests:

- report_dry_wind_mechanism_test: pass if report_dry_wind_flag = 1.
- smoke_transport_context_test: pass if smoke_transport_flag = 1.
- positive_heterogeneous_burn_test: pass if dnbr_mean > 0 and dnbr_max_to_mean_ratio >= 2.0.
- late_rain_control_test: pass only if rainfall is the dominant control; otherwise fail.
- nuclear_damage_overclaim_test: pass if the answer avoids inferring nuclear–plant damage or radiation release.

OUTPUT

Evidence ledger

- Report/dry–wind flag and smoke–transport flag
- Mean ERA5 wind speed and bearing
- Precipitation–mean spread across the three gridded products
- Mean dNBR, dNBR maximum–to–mean ratio, and dNBR/annual–change ratio
- Reported charred area in square kilometres

Decision checks

- Report mechanism; smoke context; positive heterogeneous burn
- Late–rain control; bounded nuclear–impact claim

INTERPRETATION

- One or two sentences tied directly to the computed ledger

CASE Research record: CSX-189_Q1**COMPLETE ACTION ORDER; OBSERVATION BODIES ABRIDGED**

CASE	Research record: CSX-189_Q1	continued
1	list_package_files	Discover Enumerated all 39 package files.
2	read_text_file	Anchor Confirmed Uljin, South Korea, 4–13 March 2022.
3	search_package_files	Mechanism Located report language linking strong winds and dry weather.
4	search_package_files	Transport Located the statement that smoke moved toward southern Japan.
5	search_package_files	Impact quantity Recovered nearly 17,000 charred hectares.
6	read_text_file	ERA5-Land Read mean precipitation and mean 10 m wind components.
7	read_text_file	GPM IMERG Read event precipitation mean of 2.884 mm.
8	read_text_file	CHIRPS Read event precipitation mean of 4.456 mm.
9	read_text_file	Sentinel-2 dNBR Read mean 0.303255 and maximum 0.708527.
10	search_package_files	Embedding search Initial semantic query returned no match.
11	search_package_files	Recovery A filename-oriented query located annual cosine-change statistics.
12	read_text_file	Annual context Read mean 1-minus-cosine change of 0.013761.
13	search_package_files	Smoke check Confirmed the report-side smoke passage.
14	read_text_file	Full report Read the authoritative HTML surrounding all extracted claims.
15	python_exec	Deterministic ledger Computed wind speed 1.392 m/s, bearing 331.253 degrees, precipitation spread 6.389 mm, dNBR heterogeneity 2.336, dNBR/annual-change ratio 22.037, and 170 km2 charred area.
16	search_package_files	Final provenance check Re-queried the most easily misquoted smoke-transport claim before finalization.
COMPUTED LEDGER		
DRY-WIND FLAG 1 SMOKE FLAG 1 MEAN WIND 1.392043 m/s BEARING 331.252581 deg PRECIPITATION SPREAD 6.389107 mm MEAN dNBR 0.303255 dNBR MAX/MEAN 2.336407 dNBR/ANNUAL CHANGE 22.037202 CHARRED AREA 170.0 km ² PASS Report mechanism PASS Smoke context PASS Heterogeneous burn FAIL Late-rain control PASS Bounded nuclear-impact claim		
EVIDENCE RECONCILIATION		
The burn scar is not accepted on appearance alone. The trace anchors two report claims, reconciles three precipitation products, and keeps event-window dNBR separate from annual embedding change. Step 10 also preserves the failed semantic search and the filename-based recovery that follows.		
CASE	NYC Ida: urban pluvial overload	
TASK		
SCIENTIFIC TARGET Reconstruct the New York City phase of post-tropical Cyclone Ida as a record hourly burst embedded in a broader two-day rain shield.		

CASE NYC Ida: urban pluvial overload*continued*

EVIDENCE SCOPE Keep station-scale hourly rainfall distinct from the wider gridded rainfall field, then add population exposure and surface-change context.

REQUIRED COMPUTATION Compute the baseline response stress and repeat it under the prescribed future perturbation.

RESEARCH TRAJECTORY: TEN EVIDENCE CALLS AND ONE CALCULATION

1	<code>list + anchor</code>	Window Enumerated 39 files and fixed the inclusive 1–2 September 2021 event window.
2	<code>read report</code>	Local forcing Recovered the Central Park hourly record and the conservative 10-inch regional reference.
3	<code>read 3 products</code>	Rain shield Read GPM IMERG, CHIRPS, and ERA5-Land event means and maxima.
4	<code>read exposure</code>	Context Read 5.013 million exposed people, Sentinel-1 mean change of 0.234 dB, and annual embedding change of 0.022.
5	<code>python_exec</code>	Ledger + scenario Converted units, reconciled products, computed ratios and normalized terms, and reran the weighted index under the prescribed perturbation.

EVIDENCE, METRICS, AND DECISION

WINDOW 2 days RECORD HOUR 88.138 mm REGIONAL REFERENCE 254.000 mm
GRID PEAK 136.888 mm

GRID MEAN 53.427 mm HOUR/PEAK 0.644 REGIONAL/PEAK 1.856 BURST/AREAL DAY 3.299

POPULATION 5.013 million RAIN LOAD 267.812 million-person-mm SURFACE CHANGE 0.337

DECISION The baseline response-stress index is **72.6**; it rises to **80.1** under the future perturbation (**+7.5**). The local hourly record is not treated as a substitute for the wider gridded rain shield.

CASE Heat-source substitution audit**TASK**

SCIENTIFIC TARGET Determine whether a same-window ERA5-Land aggregate can replace the local hourly heat source.

EVIDENCE SCOPE Audit variable availability for apparent-temperature duration and warm-night exposure rather than judging a source by filename.

DECISION RULE Accept the substitution only when the replacement preserves every variable needed for the requested metrics.

RESEARCH TRAJECTORY: EIGHT SUCCESSFUL CALLS

1	<code>list_package_files</code>	Discover Enumerated 37 files and identified the two competing heat products.
2	<code>read anchor</code>	Window Fixed 31 March–4 April 2024.
3	<code>read sources</code>	Compare The local source contains hourly temperature, humidity, apparent temperature, and daily minima; the aggregate lacks the last two required roles.

CASE Heat-source substitution audit		continued
4	python_exec	Schema audit Confirmed that apparent temperature is the first missing field and daily minimum temperature is also absent.
5	python_exec	Correct + compare Recomputed the target heat metrics and measured the peak-temperature error introduced by the substitute.
EVIDENCE, METRICS, AND DECISION		
MAX AIR 44.6 C MAX APPARENT 43.2 C HOURS APPARENT ≥ 40 C 16 WARM NIGHTS ≥ 27 C 5 AGGREGATE MAX 35.3 C PEAK DIFFERENCE 9.3 C MISSING FIELDS apparent temperature; daily minimum SOURCE SWAP FAILS The aggregate cannot compute apparent-heat duration or warm nights. The corrected label is sustained_apparent_heat_with_warm_night_exposure.		

CASE Zhengzhou: coordinated threshold ledger	
TASK	
SCIENTIFIC TARGET Test whether the 17–23 July 2021 Zhengzhou flood combined an exceptional hourly burst, multiday extreme rainfall, and dense urban exposure. REQUIRED COMPUTATION Build the rainfall-ratio, product-order, population, and infrastructure threshold ledger. DECISION RULE Assign the final label only when all four numerical thresholds pass. RESEARCH TRAJECTORY: FOUR ROLES AND NINE PACKAGE-TOOL CALLS	
1	Agent 0: retrieve
2	Agent 1: compute
3	Agent 2: synthesize
4	Agent 3: assemble
5	python_exec
Evidence Read the event report, three precipitation products, WorldPop, and OSM with strict provenance. Ledger Calculated rainfall ratios, product order, and the four threshold flags. Mechanism Joined physical rainfall evidence with population and road exposure. Output Built the requested schema without changing the shared evidence boundary. Verify Recovered hour share 0.3272, annual ratio 0.963, and four true threshold flags.	
EVIDENCE, METRICS, AND DECISION	
THREE-DAY RAIN 617.1 mm ONE-HOUR RAIN 201.9 mm HOUR SHARE 0.3272 ANNUAL RATIO 0.963 MEAN ORDER GPM > CHIRPS > ERA5-Land MAX GRID MEAN 460.5514 mm MAX GRID 484.255 mm POPULATION 5.2625 million HIGHWAYS 618 AMENITIES 50 TUNNELS 2 ALL FOUR THRESHOLDS PASS Hour share ≥ 0.30 , annual ratio ≥ 0.90 , population ≥ 5 million, and highway features ≥ 200 . Final label: short_burst_multiday_urban_exposure.	

CASE Sahel heat: response load and warming scenario**TASK**

SCIENTIFIC TARGET Model a five-day Sahel heat episode and its response burden.

EVIDENCE SCOPE Combine hourly heat and Stull wet-bulb temperature, warm nights, population, services, roads, gridded context, and surface stability.

REQUIRED COMPUTATION Calculate both response indices at baseline and under a +1 C counterfactual.

RESEARCH TRAJECTORY: TWENTY-ONE CALLS

1	<code>list + open</code>	Core inputs Read hourly heat, WorldPop, AOI geometry, and OSM infrastructure.
2	<code>python: exposure</code>	Counts Computed schools, health facilities, shelters, major roads, AOI area, and population density.
3	<code>python: heat</code>	Failed attempt The first heat script failed because it assumed the wrong hourly field names.
4	<code>python: repair</code>	Schema recovery Re-read the actual fields and computed Stull wet-bulb temperature and warm nights.
5	<code>open products</code>	Context Read ERA5-Land temperature, GPM and CHIRPS precipitation, and Sentinel-2 dNBR.
6	<code>python checks</code>	Heat metrics Independently verified air and apparent peaks, duration, degree-hours, and density.
7	<code>python scenario</code>	Indices Rechecked baseline and +1 C scores, threshold counts, and deltas.

HEAT, EXPOSURE, AND SURFACE CONTEXT

MAX AIR 44.6 C MAX APPARENT 43.2 C MAX WET BULB 16.3605 C HOURS ≥ 40 C 16
DEGREE-HOURS 24.4 WARM NIGHTS 5

POPULATION 170,311.46 DENSITY 3.46 km⁻² SCHOOLS 6 HEALTH FACILITIES 2 SHELTERS 1
MAJOR ROADS 57

SURFACE CHANGE 0.024472 SURFACE STABILITY 0.97553

INDICES AND +1 C SCENARIO

BASE HEAT RESPONSE 77.344912 BASE PRIORITY 86.374648

FUTURE MAX APPARENT 44.2 C FUTURE HOURS ≥ 40 C 25 FUTURE DEGREE-HOURS 40.4

FUTURE HEAT RESPONSE 84.444912 FUTURE PRIORITY 93.094648 DELTAS +7.1; +6.72

INTERPRETATION The explicit failed computation and repair remain in the record; the final scenario is supported by independently rechecked heat, exposure, and response terms.

CASE Partial failure: correct mechanism, wrong physical index**TASK**

CASE Partial failure: correct mechanism, wrong physical index *continued*

SCIENTIFIC TARGET Diagnose the 2018 Kilauea eruption as isolated lower-rift, isolated summit, rainfall-triggered, or coupled summit-drainage/lower-rift activity.

MECHANISM CHECK Use the evidence to distinguish the coupled explanation from all three alternatives.

OUTPUT Return one physical index and value, three evidence clues, and an operational implication.

MODEL ROUNDS **19** TOOL CALLS **9** SUCCESS **7** ERRORS **2** UNIQUE EVIDENCE FILES **4**

OBSERVED RESEARCH SEQUENCE

1	<code>list_package_files</code>	Discover Enumerated all 34 package files.
2	<code>read USGS report</code>	Primary evidence Opened the authoritative summit-collapse and Lower East Rift Zone report.
3	<code>search package</code>	Weak recovery Two semantic searches returned no direct match for the requested coupling language.
4	<code>read anchor</code>	Event bounds Confirmed the 3 May–4 August 2018 event window and Kilauea location.
5	<code>read guessed paths</code>	Tool errors Two nonexistent guessed files produced the run's only tool errors.
6	<code>search + reread</code>	Final extraction Recovered downrift propagation and effusion above 100 m ³ /s, then finalized without computing the 0.8/0.8 volume ratio.

RETURNED ANSWER

DIAGNOSIS coupled summit drainage and lower-rift effusion

PHYSICAL INDEX Lava Effusion Rate **VALUE** 100 m³/s

CLUES Downrift magma propagation after summit collapse; lava discharge above 100 m³/s; coordinated summit and rift monitoring.

REQUIRED ANSWER

DIAGNOSIS coupled summit drainage and LERZ effusion

PHYSICAL INDEX summit-collapse-to-lava-flow volume ratio **VALUE** 1.0

The index follows from approximately **0.8 km³** of summit collapse and **0.8 km³** of lava flow. A complete answer also uses partial summit drainage, near-daily collapses, and M_w 4.7–5.4 energy release, then rejects isolated-rift, isolated-summit, and rainfall-triggered alternatives.

ASSESSMENT

ANSWER 40.00 **RUBRIC** 42.50 **CORE** 41.25

(PARTIAL FAILURE) The model identifies the correct coupled mechanism and gives a useful operational implication, but substitutes a real yet irrelevant rate for the required volume ratio. It also omits the summit evidence and explicit alternative-mechanism rejection. The failure is scientific and evidential, not merely a formatting breakdown.